

KENYATTA UNIVERSITY

BACHELOR OF Ed. Sc. PROGRAMME

Department of Physics

SPH 290: Mechanics II

2023/2024 SEM 1

B.Ed -Lecturer

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Lessons to be covered

- There are six Lessons to be Covered in this Module.
- For each lesson, you are provided “**key Learning Outcomes**”, telling you what you should be able to do/achieve. – **Read them carefully** in the module.
- The ONE HOUR LECTURES, will only focus on the ground breaking of various parts of a Lesson.

LESSON 1

INTRODUCTION TO THE MECHANICS OF ROTATIONAL MOTION

LEARNING OUTCOMES

(**Provided elsewhere** – ensure you understand them)

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1.1 Overview of Mechanics II

Mechanics (in general)- study of the effects of interaction between force and a body.

Effects can be:

- Change of state – **deformation** or **motion**(linear or angular).
- No change of state (Dynamic or static) – equilibrium

The study of Mechanics is divided into three:

- **Kinematics** - describing **motion** (in Equation).
- **Dynamics** –how to **change** the state of a body .

- **Equilibrium** - conditions for **stability of state** of a body

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1.0 Introduction

Mechanics II deals with **Rotational Motion** (Angular Motion) of **Rigid bodies**.

Definition:

A rigid body is one whose **particles remain fixed** in one Position, relative to each other, within the body, even in motion.

Why study Rotational Motion?

Our **everyday life** is full of / depends on rotational motion of **rigid bodies**.

Machines like: bicycles, vehicles, industrial machines
galaxies, orbiting planets, among others

1.1. Description of Rotational Motion

Every form of motion is characterized by the behavior of quantities such as:

- **displacement**
- **velocity and**
- **Acceleration**

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Uniform or non-uniform and related **by Equations .**

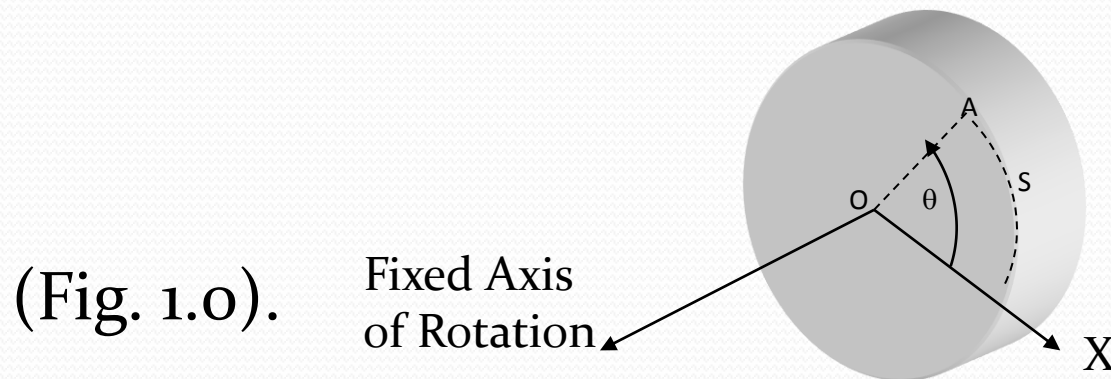
For **Rotational motion**, these quantities are to be given the nametag “**angular**”, as:

- Angular **displacement**
- Angular **velocity**
- Angular **acceleration**

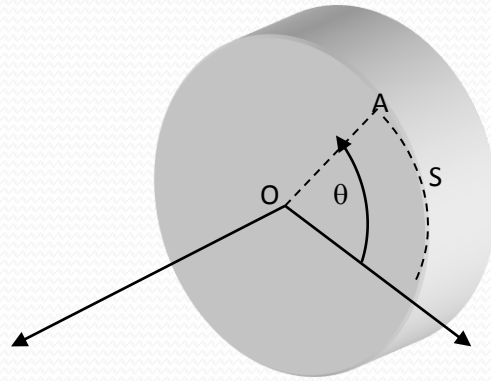
Thus, the rotational motion of a rigid body is described by:
angular displacement of a **reference point** or **line** ,
about a fixed axis (Axis of Rotation).

1.3 Angular Displacement

Consider a displacement of a point like A or a line OA on a disc, from a reference line OX, by rotation about a fixed perpendicular ($\perp$) axis, in the anticlockwise direction



The **displacement angle θ** is representative of the **angular displacement** of every point on the body/whole body.



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the linear displacement (s) of any point (e.g. A), along a path (s), is related to θ in a way, given by:

$$S = \frac{\theta}{360} \times 2\pi r$$

where $r = OA$

This can be written in a short form, in units called radians.

Where $360^\circ = 2\pi \text{ rad.}$, ($2\pi \text{ rad} = 1 \text{ rev}$, $\pi = 3.14$).

Thus, $S = \frac{\theta (\text{rad})}{2\pi (\text{rad})} \times 2\pi r (\text{m})$ reducing to $S = r\theta$

Equations of motion

The parameters which characterize the three “**equations of linear motion**”,

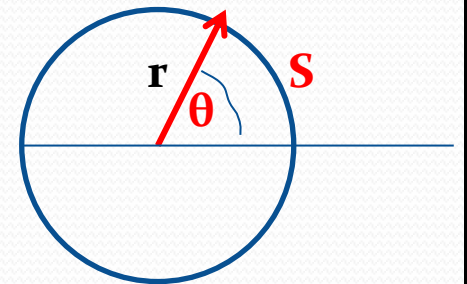
- $v = u + at$
- $v^2 = u^2 + 2as$
- $s = ut + \frac{1}{2}at^2$

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can be **transformed** into **corresponding parameters** and **equations** for rotational motion, starting from

Linear -**angular** displacement relation

$$s = r\theta$$



We can then obtain expressions relating :

- **linear velocity** or **acceleration** to their **angular equivalents**.

1.3.2. Angular velocity – **from Linear velocity**

From, $v = \frac{\text{linear Displacement}}{\text{time}} = \frac{\Delta s}{\Delta t} \approx \frac{ds}{dt}$

If we let $v = \frac{ds}{dt}$ and using **$s = r\theta$**

$$v = \frac{d(r\theta)}{dt}$$

since **r is constant** as θ changes, let $v = r \left(\frac{d\theta}{dt} \right)$

Since, $\frac{d\theta}{dt} \approx \frac{\Delta\theta}{\Delta t}$ is rate of change of **angular displacement**,

it is called **angular velocity (ω)**.

So, let $\mathbf{v} = r \left(\frac{d\theta}{dt} \right) = \mathbf{r}\omega$

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Hence, we have the following two useful relations:

- $\mathbf{v} = \mathbf{r}\omega$ (m) (velocity transformation relation)
- $\omega = \frac{d\theta}{dt}$ (rad/s) (angular velocity relation)

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1.3.3. Angular Acceleration – **from Linear acceleration**

Angular acceleration can also be obtained from linear (a):

$$a = \frac{\text{change in linear velocity}}{\text{time}} = \frac{\Delta v}{\Delta t} \approx \frac{dv}{dt}$$

Now, since, **$v = r\omega$**

Let $a = \frac{dv}{dt} = \frac{d(r\omega)}{dt}$ remember, r – **is constant**

So, let $a = \frac{d(r\omega)}{dt} = r \frac{d\omega}{dt}$ and $\frac{d\omega}{dt}$ is **rate of change of vel.**

Let $\frac{d\omega}{dt} = \alpha$, hence, **$a = r \frac{d\omega}{dt} = r\alpha$**

So, **$a = r\alpha$**

So, again we have :

- $\mathbf{a} = r\alpha$ (m/s²) (acceleration transformation relation)
- $\alpha = \frac{d\omega}{dt}$ rad/s² (angular acceleration)

The obtained relations can be applied in various problems.

Example 1.0 (a)

A wheel of a car of radius 50cm rolled on the ground covering a length of 100 m in its displacement. Calculate its angular displacement.



Solution

This is a case of relating **linear displacement** to **angular**

From, $s = r\theta$ where $r = 0.5$ m, $s = 100$ m

$$\theta = \frac{s}{r}$$

$$\theta = \frac{100}{0.5} = 200 \text{ rad}$$

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Example 1.0 (b)

If the angular position θ of a reference line on a rotating wheel is dependent on time t as

$$\theta(t) = 2t^3 - 2t + 5$$

where t is in seconds and θ is in radians, determine the expressions for:

- (i) angular velocity $\omega(t)$ and
- (ii) angular acceleration $\alpha(t)$.

Solutions

i). Angular velocity $\omega(t)$

Given **angular displacement**, as a function of time (t),

Angular velocity is given as $\omega = \frac{d\theta}{dt}$

Thus, $\omega = \frac{d}{dt}(2t^3 - 2t + 5)$

$$\omega(t) = 6t - 2$$

Thus, angular velocity is also **time dependent**.

e.g, for **t = 0**, $\omega(0) = -2 \text{ rad/sec}$

for **t = 1**, $\omega(1) = 6 - 2 = 4 \text{ rad/sec}$

(ii) Angular acceleration α (t).

Angular acceleration is given as $\alpha = \frac{d\omega}{dt}$

but **angular velocity**, $\omega(t)$, as a function of time (t),
is obtained in (i) as $\omega(t) = 6t - 2$

Therefore, $\alpha = \frac{d}{dt} (6t - 2)$

$$\alpha = 6 \text{ rad/s}^2$$

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Also, using the **transformation expressions**, the earlier stated **Eqs. of linear motion** can be transformed into **equations of rotational motion**.

By substitution:

- i). $v = u + at$ becomes $r\omega = r\omega_o + (r\alpha)t$
hence, $\omega = \omega_o + \alpha t$
- ii) $v^2 = u^2 + 2as$ becomes $(r\omega)^2 = (r\omega_o)^2 + 2(r\alpha)(r\theta)$
hence, $\omega^2 = \omega_o^2 + 2\alpha\theta$
- iii) $s = ut + \frac{1}{2}at^2$ becomes $\theta = \omega_o t + \frac{1}{2}\alpha t^2$

1.4.1 Applications of the Equations of rotational Motion

The **equations** are used the same way those of linear motion are used, in this case, to determine any of **the parameters** that characterize **rotational motion**.

Example

- 1) The wheel of a car starts its rotation **from rest** at some reference point. If it has a constant angular acceleration of **5 rad/s^2** , calculate its;
 - angular displacement after 10 seconds.
 - angular velocity at the 10th second.

Solutions

From the list of given/known quantities

$$\omega = ?$$

$$\omega_0 = 0$$

$$t = 10 \text{ sec}$$

$$\alpha = 5 \text{ rad/s}^2$$

$$\theta = ?$$

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i). The angular displacement θ ,

Using $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$

$$\theta = 0 + \frac{1}{2} \times 5 \times 10^2$$

thus, $\theta = 250 \text{ rad}$

Alternatively, in revolutions (rev),

- From, $2\pi \text{ rad} = 1 \text{ rev.}$, therefore, $\theta = 40 \text{ rev}$

ii). Angular velocity attained at the 10th second,

- Meaning at, $t = 10 \text{ sec}$

Let the angular velocity be $= \omega$

Therefore, from $\omega = \omega_0 + \alpha t$

$$\omega = 0 + 5 \times 10$$

hence, $\omega = 50 \text{ rad/s}$

Alternatively, in revolutions (rev/s),

From, $2\pi \text{ rad} = 1 \text{ rev}$

$$50 \text{ rad/s} = \frac{50 \text{ rad/s} \times 1 \text{ rev}}{2\pi \text{ rad}}$$

thus, $\omega = 8 \text{ rev/s}$

Example

2) If a similar wheel having a velocity of 12 rad/s starts to slow down at a constant retardation of 1.5 rad/s². How long does it take to come to rest?

The information given means

$$\omega = 0$$

$$\omega_o = 12 \text{ rad/s}$$

$$t = ?$$

$$\alpha = -1.5 \text{ rad/s}^2 \quad (-\text{ve for retardation})$$

From $\omega = \omega_o + \alpha t$

$$0 = 12 - 1.5 t$$

Therefore, $t = (12/1.5) = 8 \text{ sec}$

E-tivity 1.0

Provided in the Module

Ensure :

- you read and understand what to be done.
- You do the stated e-tivities.



LESSON 2

VECTORS AND PROPERTIES

LEARNING OUTCOMES

(**Provided elsewhere** – ensure you understand them)

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2.0. Introduction

In the study of Physics in general, we work with **scalar** and **vector** quantities

Of interest here is that, **vector** quantities may **yield a scalar** or another **vector quantity**, **depending** on the **mathematical** operation.

As a result, we need to know what **mathematical operators** produce either **scalar** or **vector**.

2.1. Dot and cross products

We introduce two mathematical operators, which are **Multiplication operators** with different rules. They are:

- **Cross product** $(\times)$ - leads to **vector** quantities
- **Dot product** $(\bullet)$ - leads to **scalar** quantities

That is, given **two vectors A and B** the products are expressed as:

$$\mathbf{A} \times \mathbf{B} = \mathbf{C} \quad (\text{vector output})$$

$$\mathbf{A} \bullet \mathbf{B} = C \quad (\text{scalar output})$$

To understand the **rules for evaluating** each of the products $\mathbf{A} \times \mathbf{B}$ and $\mathbf{A} \bullet \mathbf{B}$, we need vectors in their components form.

- Every vector has components , which may be **non-zero** or **zero** along the **x**, **y** and **z**-axes.
- Each vector can be expressed in terms of **unit vectors** (**i**, **j**, **k**) associated with values along the axes, respectively, in the form:

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$$

$$\mathbf{B} = B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}$$

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Definition:

A unit vector is a vector whose magnitude is a unit.

For any vector given as $\mathbf{A} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$, its magnitude is expressed as:

$$|\mathbf{A}| = \sqrt{x^2 + y^2 + z^2}$$

We know that when **a value** is divided by its **magnitude** we get **a unit** (1).

Therefore, similarly, **a unit vector** is obtained when **a vector** is **divide by its magnitude**.

i.e **a unit vector** of **vector** $\mathbf{A}$ would be expressed as:

$$\hat{\mathbf{A}} = \frac{\mathbf{A}}{|\mathbf{A}|}$$

Example:

Given a vector as $\mathbf{A} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$, the unit vector of this vector would be:

$$\text{From } \hat{\mathbf{A}} = \frac{\mathbf{A}}{|\mathbf{A}|}$$

$$\hat{\mathbf{A}} = \frac{x \mathbf{i} + y \mathbf{j} + z \mathbf{k}}{\sqrt{x^2 + y^2 + z^2}}$$

$$= \frac{x \mathbf{i}}{\sqrt{x^2 + y^2 + z^2}} + \frac{y \mathbf{j}}{\sqrt{x^2 + y^2 + z^2}} + \frac{z \mathbf{k}}{\sqrt{x^2 + y^2 + z^2}}$$

Along each axis, values for other axes are to be zero, such that

$$\begin{aligned}\hat{\mathbf{A}} &= \frac{x \mathbf{i}}{\sqrt{x^2 + y^2 + z^2}} + \frac{y \mathbf{j}}{\sqrt{x^2 + y^2 + z^2}} + \frac{z \mathbf{k}}{\sqrt{x^2 + y^2 + z^2}} \\ &= \frac{x \mathbf{i}}{\sqrt{x^2}} + \frac{y \mathbf{j}}{\sqrt{y^2}} + \frac{z \mathbf{k}}{\sqrt{z^2}}\end{aligned}$$

Hence, $\hat{\mathbf{A}} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

Thus, $\mathbf{i}$ - is unit vector of the vector $\mathbf{A}$ along x-axis

$\mathbf{j}$ - is unit vector of the vector $\mathbf{A}$ along y-axis and

$\mathbf{k}$ - is unit vector of the vector $\mathbf{A}$ along z-axis

2.1.1. Evaluation of Cross product

There are a few **rules to learn** and **a few to prove**.

For example, **cross product** can be evaluated by **multiplication of brackets**, but it is easier done by **a matrix method**.

While the **two methods** **give the same results**, the **multiplication of brackets** requires **knowledge of certain properties** of the products of the **unit vectors** i , j and k .

a). Method of multiplication of brackets

let $\mathbf{A} \times \mathbf{B} = (A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}) \times (\mathbf{B}_x \mathbf{i} + \mathbf{B}_y \mathbf{j} + \mathbf{B}_z \mathbf{k})$

Expanding as per the normal multiplication process;

$$\text{Let } \mathbf{A} \times \mathbf{B} = A_x \mathbf{i} (\mathbf{B}_x \mathbf{i} + \mathbf{B}_y \mathbf{j} + \mathbf{B}_z \mathbf{k}) + A_y \mathbf{j} (\mathbf{B}_x \mathbf{i} + \mathbf{B}_y \mathbf{j} + \mathbf{B}_z \mathbf{k}) \\ + A_z \mathbf{k} (\mathbf{B}_x \mathbf{i} + \mathbf{B}_y \mathbf{j} + \mathbf{B}_z \mathbf{k})$$

On expanding the brackets, we have:

$$= A_x \mathbf{B}_x (\mathbf{i} \times \mathbf{i}) + A_x \mathbf{B}_y (\mathbf{i} \times \mathbf{j}) + A_x \mathbf{B}_z (\mathbf{i} \times \mathbf{k}) + A_y \mathbf{B}_x (\mathbf{j} \times \mathbf{i}) + A_y \mathbf{B}_y (\mathbf{j} \times \mathbf{j}) \\ + A_y \mathbf{B}_z (\mathbf{j} \times \mathbf{k}) + A_z \mathbf{B}_x (\mathbf{k} \times \mathbf{i}) + A_z \mathbf{B}_y (\mathbf{k} \times \mathbf{j}) + A_z \mathbf{B}_z (\mathbf{k} \times \mathbf{k})$$

The following are useful properties of the **cross products** of unit vectors:

if there are **two vectors** **A** and **B**

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \sin \theta \quad \text{but} \quad \mathbf{B} \times \mathbf{A} = - |\mathbf{A}| |\mathbf{B}| \sin \theta$$

A and **B** may even be **unit vectors** **i**, **j**, **k**.

- If the **angle between** **A** and **B** is $\theta = 0$ then **Sin** $\theta = 0$

Thus, the cross product of two similar or parallel vector = 0

e.g

$$\mathbf{i} \times \mathbf{i}, \mathbf{j} \times \mathbf{j}, \mathbf{k} \times \mathbf{k} = 0 \quad (\text{involve } \textbf{parallel vectors} \text{ each } \theta = 0)$$

- If the **angle between** **A** and **B** is $\theta = 90$ then **Sin** $\theta = 1$,

So, $\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}|$

i.e from $\mathbf{A} \times \mathbf{B} = \mathbf{C}$ means, $|\mathbf{A}||\mathbf{B}| \equiv \mathbf{C}$

(The resulting vector **C** is along the **axis of rotation**, $\perp$ **to plane of A and B**)

e.g

$$\mathbf{i} \times \mathbf{j} = \mathbf{k}, \quad \mathbf{k} \times \mathbf{i} = \mathbf{j}, \quad \mathbf{j} \times \mathbf{k} = \mathbf{i}, \quad \text{since } \theta = 90 \text{ and } \text{Sin } 90 = 1$$

From which **the reverse order** gives (**- ve** vector)

$$\mathbf{j} \times \mathbf{i} = -\mathbf{k},$$

$$\mathbf{i} \times \mathbf{k} = -\mathbf{j},$$

$$\mathbf{k} \times \mathbf{j} = -\mathbf{i}$$

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This cases can be proved by the **Right Hand Grasp** rule;

Where the right hand **fingers are set to point** in the **direction of the rotation**, the **outstretched thumb** then indicates the direction of the result.

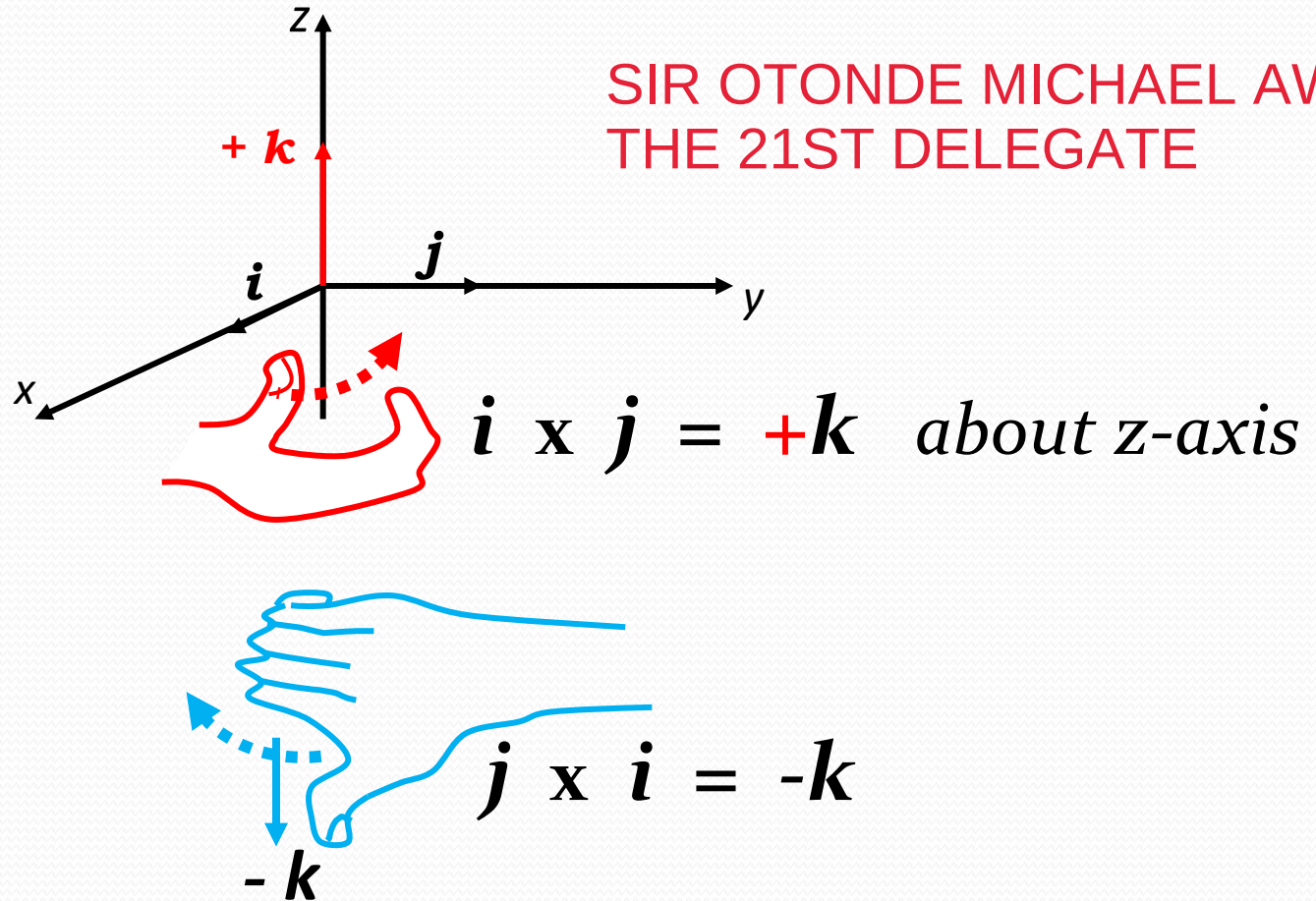
The direction of rotation is read as:

from first **unit VECTOR** towards **SECOND unit** vector , in the **order** of their **cross product** . E.g

$i \times j$ rotation from ***x-axis*** (***i***) towards ***y-axis*** (***j***)

$j \times i$ rotation from ***y-axis*** towards ***x-axis***

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The **Outstretched THUMB** is along the **AXIS** of Rotation and the **resultant unit vector** for the cross product

With these properties, the product of the two brackets

$$\text{Simplify as } \mathbf{A} \times \mathbf{B} = \mathbf{0} + A_x B_y \mathbf{k} - A_x B_z \mathbf{j} - A_y B_x \mathbf{k} + \mathbf{0} + A_y B_z \mathbf{i} \\ + A_z B_x \mathbf{j} - A_z B_y \mathbf{i} + \mathbf{0}$$

When grouped by the **unit vectors**;

$$\mathbf{A} \times \mathbf{B} = (A_y B_z \mathbf{i} - A_z B_y \mathbf{i}) + (A_z B_x \mathbf{j} - A_x B_z \mathbf{j}) + (A_x B_y \mathbf{k} - A_y B_x \mathbf{k}) \\ = (A_y B_z - A_z B_y) \mathbf{i} - (A_x B_z - A_z B_x) \mathbf{j} + (A_x B_y - A_y B_x) \mathbf{k}$$

$$\text{Thus, } \mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y) \mathbf{i} - (A_x B_z - A_z B_x) \mathbf{j} + (A_x B_y - A_y B_x) \mathbf{k}$$

Conclusion

- We notice from the result that the outcome is a vector quantity, it has components along the x , y and z axes.
- While it works, **it is tedious** and requires **knowing the expressions** for **evaluating the cross products** of the **unit vectors**

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b). Method of Matrix

The rule for the matrix method shortens the process of evaluating cross products.

If we consider the same vectors used above.

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$$

$$\mathbf{B} = B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}$$

In matrix , the expression is written as:

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} i & j & k \\ \boxed{} & A_y & A_z \\ B_x & \boxed{} & B_z \end{vmatrix} \quad \begin{vmatrix} i & j & k \\ A_x & \boxed{} & A_z \\ B_x & \boxed{} & B_z \end{vmatrix} \quad \begin{vmatrix} i & j & k \\ A_x & A_y & \boxed{} \\ B_x & B_y & \boxed{} \end{vmatrix}$$

Which is reduced to **two-by-two** matrices as:

$$\mathbf{A} \times \mathbf{B} = i \begin{vmatrix} A_y & A_z \\ B_y & B_z \end{vmatrix} - j \begin{vmatrix} A_x & A_z \\ B_x & B_z \end{vmatrix} + k \begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix}$$

$$\mathbf{A} \times \mathbf{B} = i(A_y B_z - A_z B_y) - j(A_x B_z - A_z B_x) + k(A_x B_y - A_y B_x)$$

Compare with the earlier result by multiplication of brackets

$$\mathbf{A} \times \mathbf{B} = i (A_y B_z - A_z B_y) - j (A_x B_z - A_z B_x) + k (A_x B_y - A_y B_x)$$

- This shows that

$$\mathbf{A} \times \mathbf{B} = (A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}) \times (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

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2.1.2. Vectors in the same plane

It was only stated that the product of $\mathbf{A} \times \mathbf{B}$ is a vector perpendicular to the plane on which the two vectors $\mathbf{A}$ and $\mathbf{B}$ lie.

$$\mathbf{A} \times \mathbf{B} = \mathbf{C}$$

It can be proved that the **cross product** of any two vectors **in the same plane**, is a vector perpendicular to **the plane**.

Consider $\mathbf{A}$ and $\mathbf{B}$ to be lying in the same **xy-plane**, meaning they have **x and y components** but **zero components** along the **z axis**.

Thus, let the vectors be

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + 0 \mathbf{k} \quad \text{and} \quad \mathbf{B} = B_x \mathbf{i} + B_y \mathbf{j} + 0 \mathbf{k}$$

Therefore, by matrix, $\mathbf{A} \times \mathbf{B} = \begin{vmatrix} i & j & k \\ A_x & A_y & 0 \\ B_x & B_y & 0 \end{vmatrix}$

$$\mathbf{A} \times \mathbf{B} = i \begin{vmatrix} A_y & 0 \\ B_y & 0 \end{vmatrix} - j \begin{vmatrix} A_x & 0 \\ B_x & 0 \end{vmatrix} + k \begin{vmatrix} A_x & A_y \\ B_x & B_y \end{vmatrix}$$

$$= i(0 - 0) - j(0 - 0) + k(A_x B_y - A_y B_x)$$

Hence, $\mathbf{A} \times \mathbf{B} = 0\mathbf{i} - 0\mathbf{j} + (A_x B_y - A_y B_x)\mathbf{k}$

The **x** and **y** components are zero, hence, the result is a vector **along z-axis (perpendicular to the xy-plane)**.

2.1.3. Evaluation of Dot product

Multiplication of brackets can be done for **dot product** the same way as the cross product, only **substituting** the **dot** for **cross** in every term.

- There are also properties to simplifying the multiplication of **unit vectors** (i, j, k) by dot product.

For the **dot product**;

$$\mathbf{A} \bullet \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos \theta$$

- When angle between the two vectors $\theta = 0$, i.e $\mathbf{A} \parallel \mathbf{B}$,
 $\cos 0 = 1$

Hence, $\mathbf{A} \bullet \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \cos 0 = |\mathbf{A}| |\mathbf{B}|$

if $\mathbf{A}$ and $\mathbf{B}$ are parallel unit vectors, e.g $\mathbf{A} = \mathbf{i}$ and $\mathbf{B} = \mathbf{i}$

Then $|\mathbf{A}| |\mathbf{B}| = |\mathbf{i}| |\mathbf{i}| = 1$

Hence, $\mathbf{i} \bullet \mathbf{i} = \mathbf{j} \bullet \mathbf{j} = \mathbf{k} \bullet \mathbf{k} = 1$ (not a vector)

For $\theta = 90^\circ$, meaning $\mathbf{A} \perp \mathbf{B}$, hence, $\cos 90^\circ = 0$

therefore, $\mathbf{i} \bullet \mathbf{j} = \mathbf{j} \bullet \mathbf{k} = \mathbf{i} \bullet \mathbf{k} = 0$

Therefore, by expansion of the brackets, evaluating the product

$$\mathbf{A} \bullet \mathbf{B} = (A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}) \bullet (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k})$$

Let

$$\begin{aligned} \mathbf{A} \bullet \mathbf{B} &= A_x \mathbf{i} (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}) + A_y \mathbf{j} (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}) + A_z \mathbf{k} (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}) \\ &= A_x B_x (\mathbf{i} \bullet \mathbf{i}) + A_x B_y (\mathbf{i} \bullet \mathbf{j}) + A_x B_z (\mathbf{i} \bullet \mathbf{k}) + A_y B_x (\mathbf{j} \bullet \mathbf{i}) + \\ &A_y B_y (\mathbf{j} \bullet \mathbf{j}) + A_y B_z (\mathbf{j} \bullet \mathbf{k}) + A_z B_x (\mathbf{k} \bullet \mathbf{i}) + A_z B_y (\mathbf{k} \bullet \mathbf{j}) + A_z B_z (\mathbf{k} \bullet \mathbf{k}) \end{aligned}$$

Applying the properties of the dot products of unit vectors,

$$= A_x B_x + 0 + 0 + 0 + A_y B_y + 0 + 0 + 0 + A_z B_z$$

$$\mathbf{A} \bullet \mathbf{B} = A_x B_x + A_y B_y + A_z B_z \quad (\text{scalar quantity})$$

This confirms that, dot product produces a scalar quantity.

The dot product can also be done by a shorter method, as a product of column vectors.

From, $\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}$ and $\mathbf{B} = B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k}$

$$\mathbf{A} \bullet \mathbf{B} = \begin{pmatrix} A_x \mathbf{i} \\ A_y \mathbf{j} \\ A_z \mathbf{k} \end{pmatrix} \bullet \begin{pmatrix} B_x \mathbf{i} \\ B_y \mathbf{j} \\ B_z \mathbf{k} \end{pmatrix} = A_x B_x \mathbf{i} \bullet \mathbf{i} + A_y B_y \mathbf{j} \bullet \mathbf{j} + A_z B_z \mathbf{k} \bullet \mathbf{k}$$

Since $\mathbf{i} \bullet \mathbf{i} = \mathbf{j} \bullet \mathbf{j} = \mathbf{k} \bullet \mathbf{k} = 1$

$$\mathbf{A} \bullet \mathbf{B} = A_x B_x + A_y B_y + A_z B_z$$

Worked Example

Given vectors $\mathbf{A} = 4\mathbf{i} + 3\mathbf{j}$ and $\mathbf{B} = 3\mathbf{i} - 2\mathbf{k}$, determine

i). $\mathbf{A} \bullet \mathbf{B}$ and

ii). the angle between the two vectors

a). **Solution**

From
$$\mathbf{A} \bullet \mathbf{B} = \begin{pmatrix} A_x \mathbf{i} \\ A_y \mathbf{j} \\ A_z \mathbf{k} \end{pmatrix} \bullet \begin{pmatrix} B_x \mathbf{i} \\ B_y \mathbf{j} \\ B_z \mathbf{k} \end{pmatrix}$$

$$\mathbf{A} \bullet \mathbf{B} = \begin{pmatrix} 4\mathbf{i} \\ 3\mathbf{j} \\ 0\mathbf{k} \end{pmatrix} \bullet \begin{pmatrix} 3\mathbf{i} \\ 0\mathbf{j} \\ -2\mathbf{k} \end{pmatrix} = 12\mathbf{i} \bullet \mathbf{i} + 0\mathbf{j} \bullet \mathbf{j} + 0\mathbf{k} \bullet \mathbf{k}$$

$$\mathbf{A} \bullet \mathbf{B} = 12 + 0 + 0 = 12$$

b). Solution

$$\mathbf{a} \bullet \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$$

but from (a), $\mathbf{a} \bullet \mathbf{b} = 12 - 0 = 12$

hence, $12 = 18 \cos \theta$

therefore, $\cos \theta = 12/18 = 2/3 = 0.667$

hence, $\theta = 48.2$

E-tivity 2.0

Provided in the Module

Ensure :

- you read and understand what to be done.
- You do the stated e-tivities.



LESSON 3

DYNAMICS OF ROTATIONAL MOTION

LEARNING OUTCOMES

(**Provided elsewhere** – ensure you understand them)

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3.0. Introduction

Dynamics is about the **change of state** of a body-
looking at **the factors** which **influence** the change of state.

Change of state can be;

- from **rest** to **motion**,
- From **constant velocity** to **acceleration** and
- From linear to curve.

In linear motion, change of state is brought by **NET FORCE**

Expressed in **Newton's second** law; $F = ma$ (3.0)
where F – net force

Note: from $F = ma$,

- Linear **acceleration** is the parameter **signifying change**
- it depends on **net force** and **mass** (by $a = F/m$)
- Thus, it does not depend on factors like the **shape** of object.

In **rotational motion**, the change of state of a body (by **angular acceleration**) depends on:

- a quantity called **torque** (analogue **of force**),
- the **location of the axis** of rotation and
- a property called **moment of inertia** (analogue **of mass**).
- **the shape** of an object

By transformation of Equations for linear motion, we can determine **factors** which influence the **dynamics of rotational motion** and how they relate to:

- Shape
- Axis of rotation

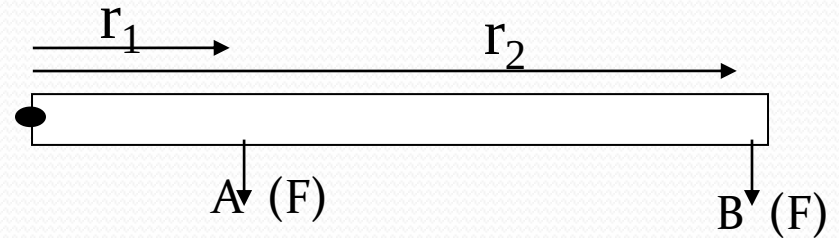
Thus, we start from what is required to **cause change**

Force (linear motion) - TO - **Torque** (Angular motion)

3.1. Torque

Torque is a term describing **how much turning effect** a given **force** can cause in **rotation**.

- It is the same as the “**moment of force**”.



Definition:

Torque is **the turning effect of force** on a body about a **fixed axis** of rotation.

- It depends on acting **force**, **distance of force from axis** and **direction** of Force.
- It is **a vector quantity**.

3.1.1. The Relation between Force and Torque

The magnitude and direction of torque (turning effect of a force)

- Is a product of force (F) and distance of force (r)
- Because it is a vector, it is cross product

In the order: $\tau = \mathbf{r} \times \mathbf{F}$

Thus, according to $\mathbf{A} \times \mathbf{B} = |\mathbf{A}| |\mathbf{B}| \sin \theta$

$$\tau = \mathbf{r} \times \mathbf{F} = |\mathbf{r}| |\mathbf{F}| \sin \theta$$

From;

$$\tau = \mathbf{r} \times \mathbf{F} = |\mathbf{r}| |\mathbf{F}| \sin \theta$$

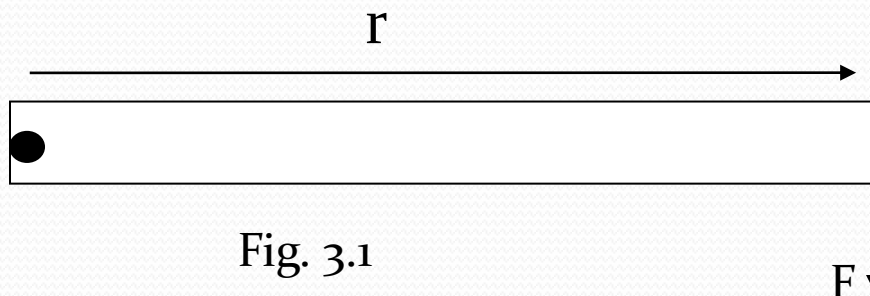
Where θ is the angle between the line of force and the distance (r).

For $\theta = 90^\circ$ as in Fig .3.1, i.e $\mathbf{r}$ and $\mathbf{F}$ are perpendicular

Hence,

$$\tau = |\mathbf{r}| |\mathbf{F}|_{\perp}$$

$$\tau = r F \text{ (Only magnitude)}$$



From the **properties of vectors**:

the cross product ($\mathbf{r} \times \mathbf{F}$) yields **a vector** along the **axis perpendicular** to the plane of $\mathbf{r}$ and $\mathbf{F}$.

Example

If $\mathbf{r}$ and $\mathbf{F}$ are on the same xy-plane, the components of their vectors would be along xy-axes but zero along z-axis.

Thus, we let $\mathbf{r} = r_x \mathbf{i} + r_y \mathbf{j} + 0\mathbf{k}$

$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j} + 0\mathbf{k}$$

Hence, from

$$\mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r_x & r_y & 0 \\ F_x & F_y & 0 \end{vmatrix}$$

On reducing to **two-by-two** matrices :

$$\begin{aligned} \boldsymbol{\tau} &= \mathbf{i} \begin{vmatrix} r_y & 0 \\ F_y & 0 \end{vmatrix} - \mathbf{j} \begin{vmatrix} r_x & 0 \\ F_x & 0 \end{vmatrix} + \mathbf{k} \begin{vmatrix} r_x & r_y \\ F_x & F_y \end{vmatrix} \\ &= \mathbf{i}(0 - 0) - \mathbf{j}(0 - 0) + \mathbf{k}(r_x F_y - r_y F_x) \end{aligned}$$

$$\boldsymbol{\tau} = 0\mathbf{i} - 0\mathbf{j} + (r_x F_y - r_y F_x)\mathbf{k}$$

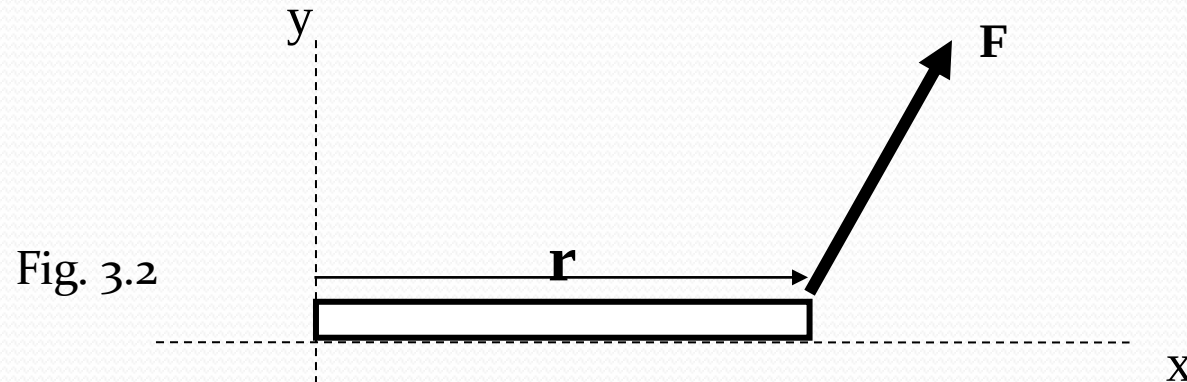
Only the component along z-axis is non-zero

$$\tau_z = (r_x F_y - r_y F_x)$$

Thus, the resulting **torque** is along **z- axis**

Example 2 (**Real physical systems**)

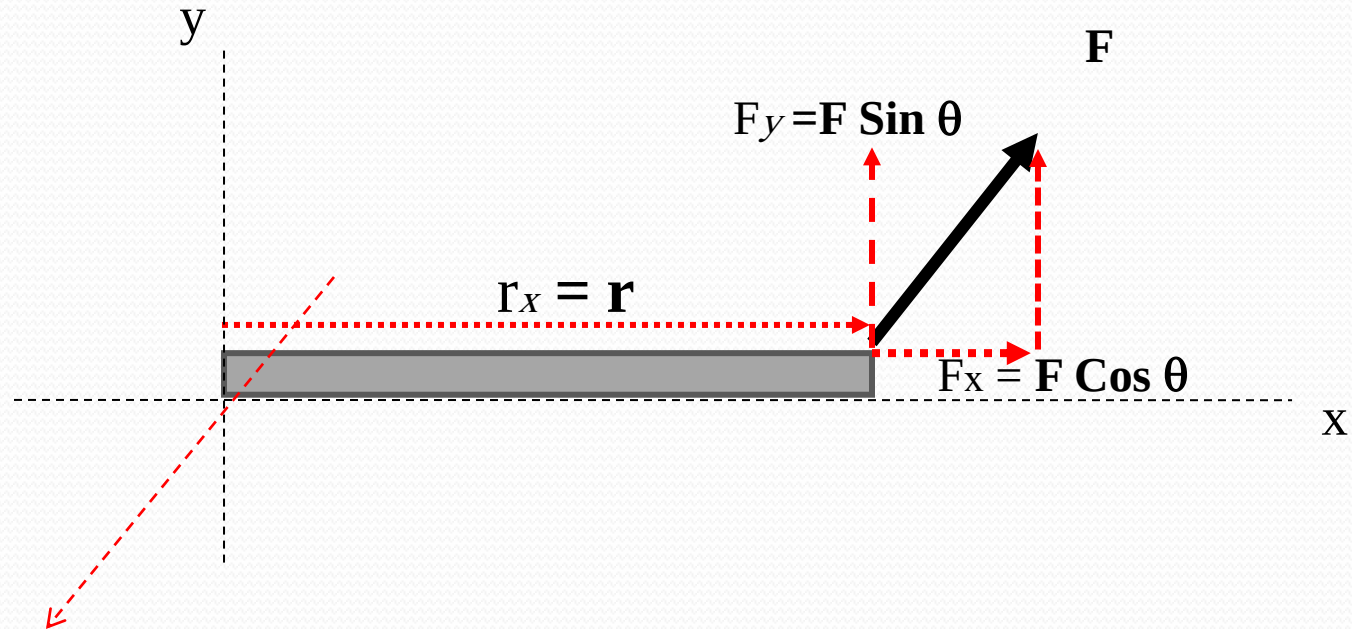
Consider force **F** acting at a distance **r** on a rod along x-axis in **xy**-plane, Fig.3.2. By expressing vectors **r** and **F** in their components, we can determine direction of **torque** along **z**.



Q. How do you determine the **moment of this force**?

Q. What is the answer?

Because $\mathbf{r}$ and $\mathbf{F}$ are in the xy -plane, they **do not have** components **along z -axis**.



Note:

- $\mathbf{r}$ is along **+ x** -axis, hence, $\mathbf{r} = \mathbf{r} \mathbf{i} + \mathbf{0j} + \mathbf{0k}$
- $\mathbf{F}$ has components along **+ x** and **+ y** axes, therefore,

$$\mathbf{F} = \mathbf{F} \cos \theta \mathbf{i} + \mathbf{F} \sin \theta \mathbf{j} + \mathbf{0} \mathbf{k}$$

Therefore, from $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$

$$\mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ r & 0 & 0 \\ F \cos \theta & F \sin \theta & 0 \end{vmatrix}$$

$$\boldsymbol{\tau} = \mathbf{i} \begin{vmatrix} 0 & 0 \\ F \sin \theta & 0 \end{vmatrix} - \mathbf{j} \begin{vmatrix} r & 0 \\ F \cos \theta & 0 \end{vmatrix} + \mathbf{k} \begin{vmatrix} r & 0 \\ F \cos \theta & F \sin \theta \end{vmatrix}$$

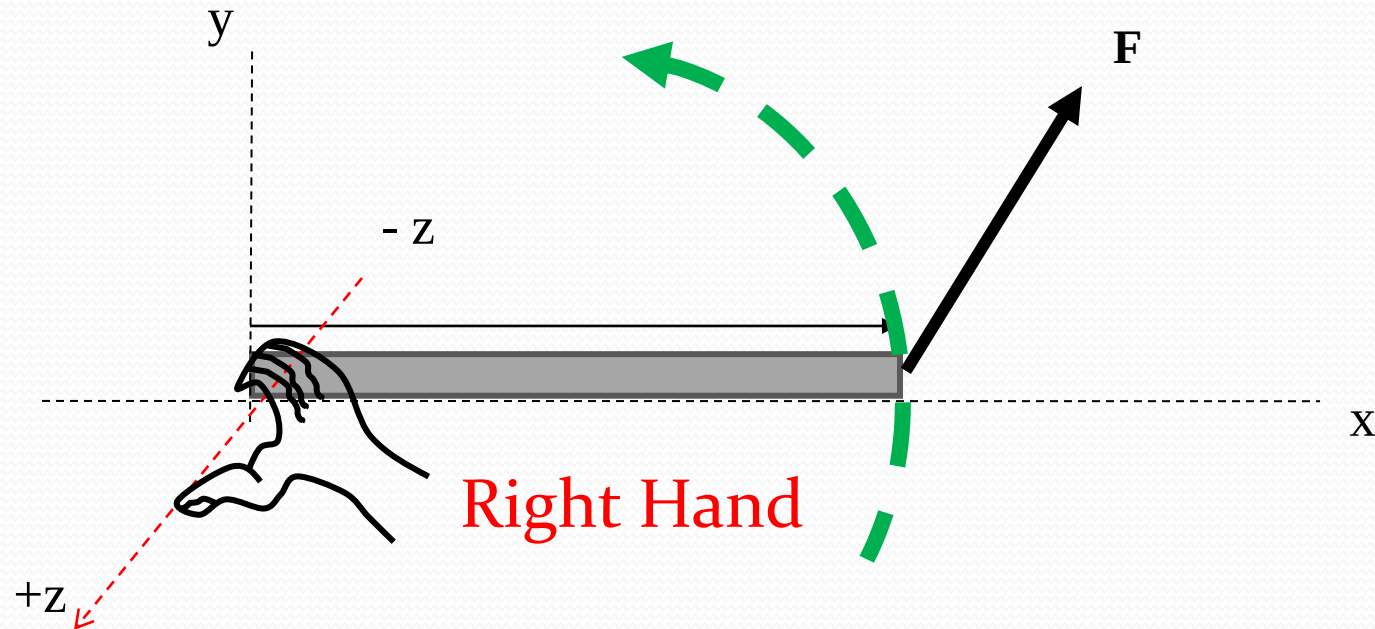
Evaluating as $= \mathbf{i}(0 - 0) - \mathbf{j}(0 - 0) + \mathbf{k}(rF \sin \theta - 0)$

Thus, $\boldsymbol{\tau} = 0\mathbf{i} - 0\mathbf{j} + rF \sin \theta \mathbf{k}$

$$\boldsymbol{\tau} = \boldsymbol{\tau}_z = rF \sin \theta$$

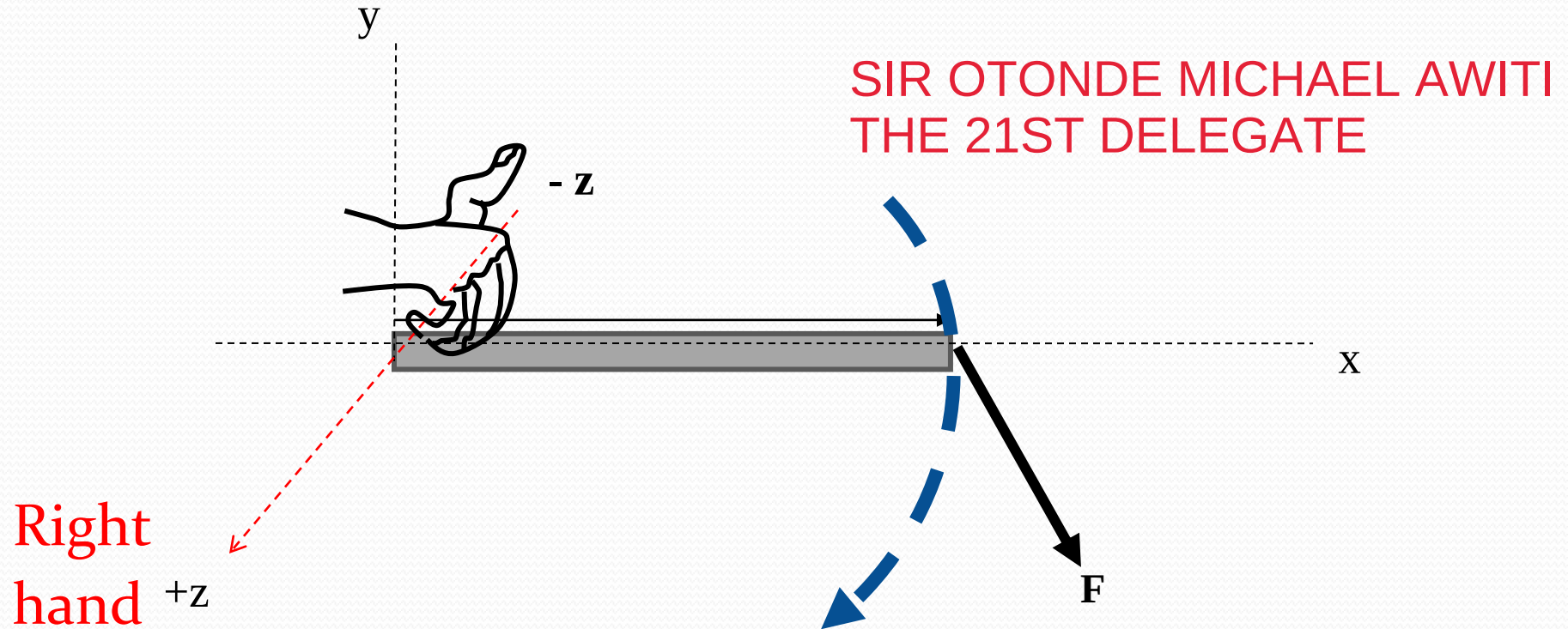
Note : it is in the **+z**–direction, due to the **direction of force**.

We can be confirmed by the **Right Hand Grasp Rule**,
(THUMB). What is the likely direction of rotation due to the
acting force (**Anticlockwise**).



If the direction of the force F were to change to **clockwise**

If the direction of the force $\mathbf{F}$ were to be as shown in Fig below.
The direction of rotation would be **clockwise**, thus, **THUMB** would
be in the **$-z$** direction.

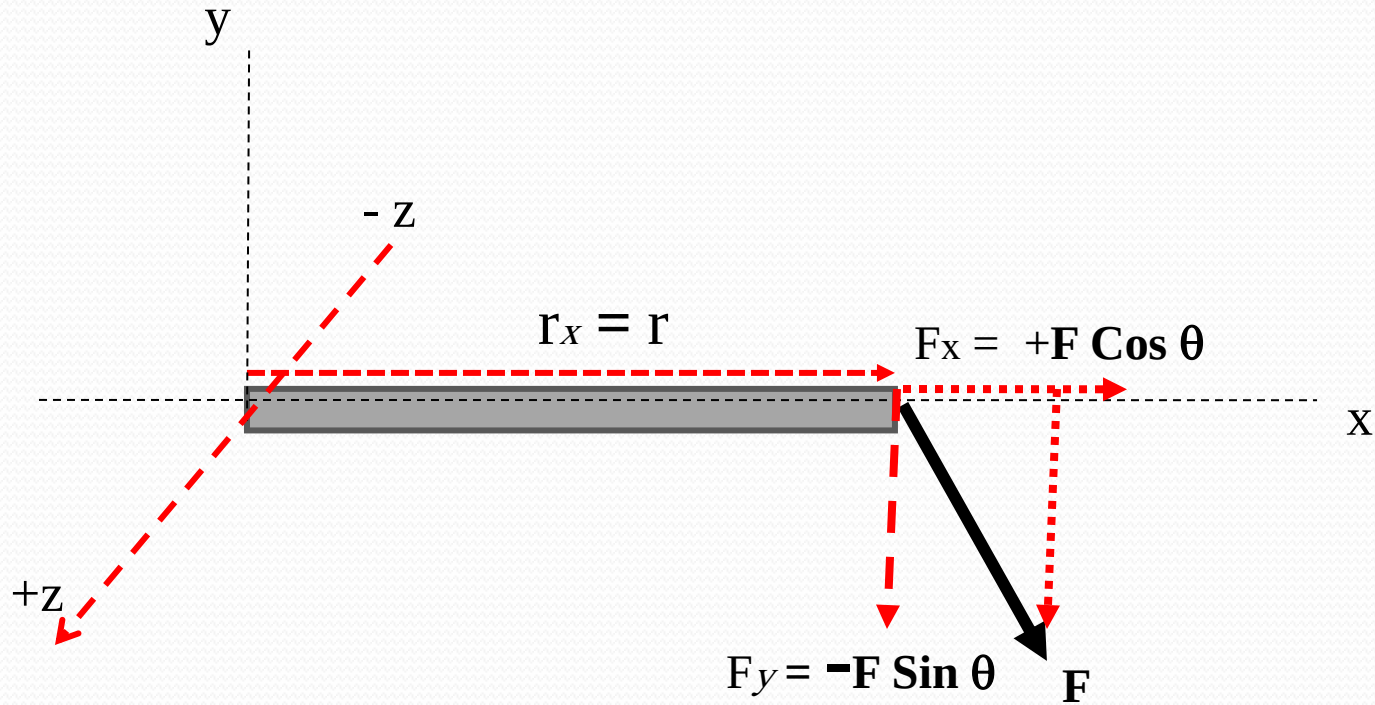


This can also be proved by **vector method**, from $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$

Note: From the fig below

- $\mathbf{r}$ is still along **+x** direction, hence, $\mathbf{r} = r \mathbf{i} + 0\mathbf{j} + 0\mathbf{k}$
- $\mathbf{F}$ has components along **+x** and **-y**, therefore,

$$\mathbf{F} = F \cos \theta \mathbf{i} - F \sin \theta \mathbf{j} + 0 \mathbf{k}$$



Therefore, from $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$

$$\mathbf{r} \times \mathbf{F} = \begin{vmatrix} i & j & k \\ r & 0 & 0 \\ FCos\theta & -FSin\theta & 0 \end{vmatrix}$$

$$\begin{aligned} \boldsymbol{\tau} &= i \begin{vmatrix} 0 & 0 \\ -FSin\theta & 0 \end{vmatrix} - j \begin{vmatrix} r & 0 \\ FCos\theta & 0 \end{vmatrix} + k \begin{vmatrix} r & 0 \\ FCos\theta & -FSin\theta \end{vmatrix} \\ &= i(0 - 0) - j(0 - 0) + k(-rFSin\theta - 0) \end{aligned}$$

Thus, $\boldsymbol{\tau} = 0i - 0j - rFSin\theta k$

It is in the negative ($-z$ direction)

$$\boldsymbol{\tau} = \tau_z = -rFSin\theta$$

3.3. Newton's 2nd law **in Rotational Motion**

In Linear Motion change of state is brought by net force as per $F = ma$ (Newton's 2nd Law)

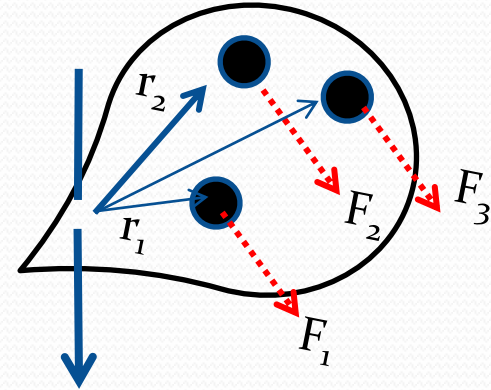
In rotational motion: -change is brought by **TORQUE**, i.e

- **Torque** is the **analogue of force** in the **change** of state.
- We need to know how (τ) relates to **angular accel** (α),
[**analogue form** for $F = ma$]

FROM relation be $\tau = r \times F$

A rigid body is composed of many particles, Fig. 3.3 .

Fig. 3.3



For a single particle
(from $\tau = \mathbf{r} \times \mathbf{F}$),

For each particle, let $\tau_i = \mathbf{r}_i \times \mathbf{F}_i = r_i F_i \sin \theta$

- Just like the **summation** of forces

$$F_1 + F_2 + F_3 + \dots = \sum F_i$$

let the **total torque** on a rigid body (due to the particles) be;

$$\tau = \tau_1 + \tau_2 + \tau_3 + \dots \sum \tau_i = \sum \mathbf{r}_i \times \mathbf{F}_i$$

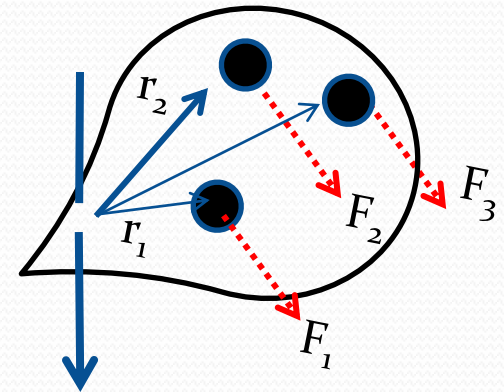
$$\text{Let } \tau = \sum (\mathbf{r}_i \times m_i \mathbf{a}_i) = \sum m_i (\mathbf{r}_i \times \mathbf{a}_i)$$

from $\mathbf{F} = m\mathbf{a}$

From the earlier relation (**linear-to- angular** ; $\mathbf{a} = r\alpha$),

$$\text{Let } \mathbf{a}_i = r_i \alpha_i$$

$$\text{So that } \sum m_i (\mathbf{r}_i \times \mathbf{a}_i) = \sum m_i r_i (\mathbf{r}_i \times \alpha_i) = \sum m_i r_i (r_i \alpha_i \sin \theta_i)$$



Thus, for case of magnitude

$$\tau = \sum m_i r_i (r_i \alpha_i \sin \theta_i) = mr^2 \alpha \sin \theta$$

For (**tangential force s**) $\theta = 90^\circ$, hence, $\sin 90 = 1$

hence, $\tau = mr^2 \alpha$ (vs **$F = ma$**)

mr^2 – is a quantity; **moment of inertia (I)** of a point mass .

If we let **$mr^2 = I$**

Then, $\tau = I \alpha \equiv$ **$F = ma$**

“I” – is the analogue quantity for mass

Definition:

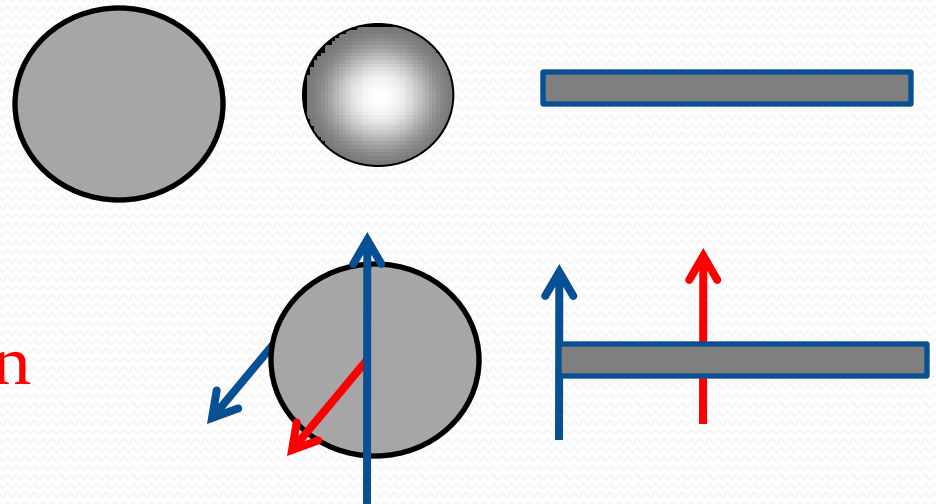
Moment of inertia of a body is **a property** which makes it **resist a change** of state in rotational motion.

$$\tau = I\alpha \qquad (\tau \propto I)$$

Moment of inertia **varies** with **a number of factors**,
unlike the mass

It varies with:

- The **shape of a body** and
- the **location** and **orientation** of the **axis** of rotation.



If the moment of inertia changes, the pattern of motion of a body changes (**smooth** or **vibrating rotation**).

- Applicable in the **alignments of wheels** of vehicles (**wheel balancing**) – to restore **alignment** and **distribution** of mass.

Problem solving requires knowledge of the moment of inertia (I) for various shapes and various axes

Here , we learn ideas of how to obtain moments of inertia for different shapes about different axes of rotation.

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Determining Moments of Inertia

There are **two methods**, depending on whether an object is **regular** or **irregular** shaped.

- **Calculation** - **summation** - discrete (**point masses**)
- **integral calculus** - **continuous mass**.
- **Experimental methods** - using inertia table (any Shape)

i). Isolated Point Masses

Recall, **the expression** $I = mr^2$

- Moment of inertia for **a point mass** of mass **m** at a distance **r** from the **axis of rotation**.

Thus, for **isolated point masses**, the moment of inertia would be the **sum of** moments of inertia

i.e if a single particle is $I_i = m_i r_i^2$

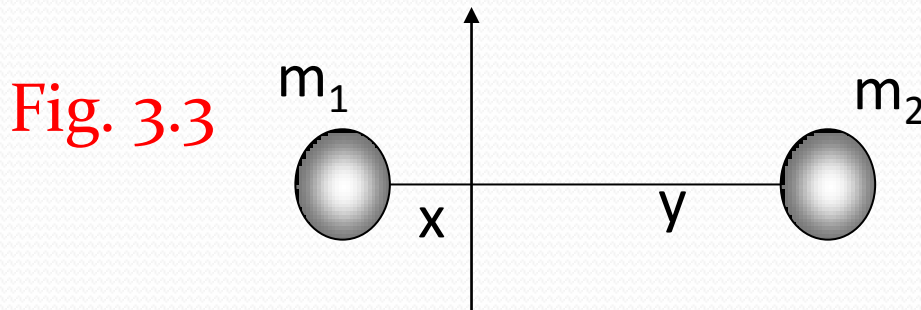
- Then, for several **isolated point masses**,

$$I = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 \dots \text{ OR}$$

$$I = \sum_{i=1}^N m_i r_i^2$$

Example

Consider a system of two point masses m_1 at a distance x and m_2 at a distance y from the axis, as in Fig 3.3.



- From
$$I = \sum_{i=1}^N m_i r_i^2$$
- The moment of inertia of this system is

$$I = m_1 x^2 + m_2 y^2$$

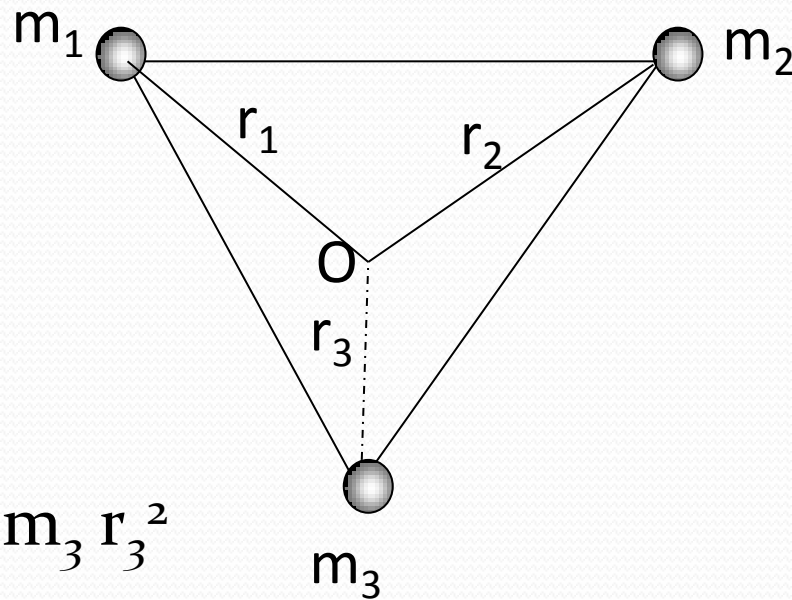
If the point masses are equal, that $m_1 = m_2 = m$, then “I” would simplify as;

$$I = m (x^2 + y^2)$$

How about a system like one in figure below, perpendicular axis through O

$$I = \sum_{i=1}^N m_i r_i^2$$

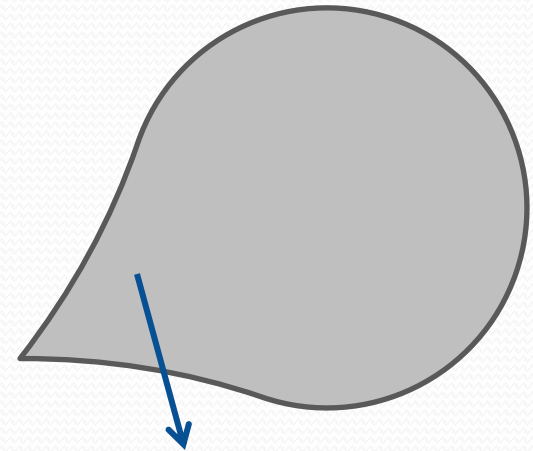
$$I = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2$$



ii). A Continuous System/mass

Fig. 3.4 is an example of a continuous distribution of mass

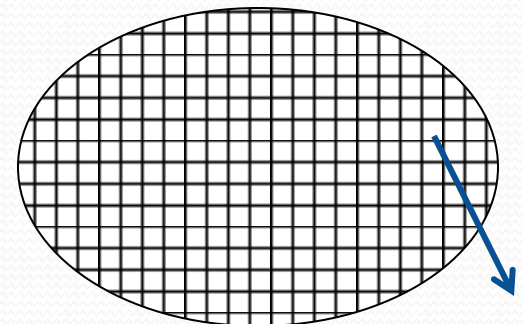
Fig. 3.4



In principle, the concept of **M.I for point mass** can be **extended** to a **continuous system of mass**.

i.e. If a body could be divided **into smaller units** (**point masses**), of equal mass, then

$$I = \sum_{i=1}^N m_i r_i^2$$



This idea is **simple in principle** but **complex in practice**.
As a result, it is replaced with a closely related method;

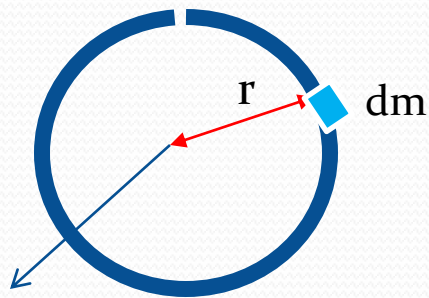
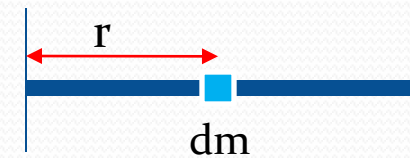
- The **Integral calculus**

This is also a **summation** of the **moments of inertia** of assumed/hypothetical building units of various shapes of mass **dm** . The pieces can be:

- Point masses
- Rectangular
- Circular rings or discs

Depending on the **shape of an object**, whether a **1D, 2D or 3D**

Assumed pieces of **point mass** work for **1D systems**;
a “**line-like object**” such as a **rod** or a **ring**, and the
expression is



$$I = \int_0^a r^2 dm$$

Where r – distance of assumed piece from the axis
 dm – assumed mass of the piece

a). A thin uniform rod (1-Dimension)

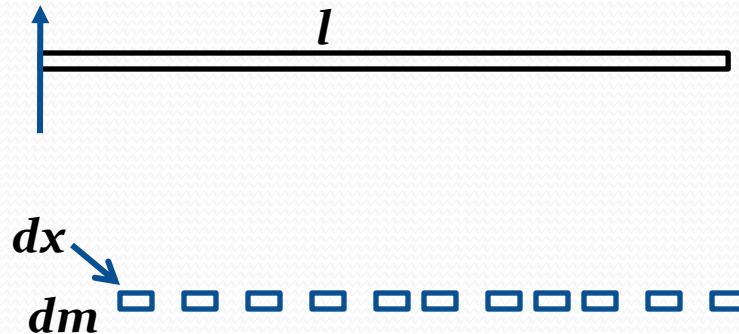
Case I: (Axis at one end)

Consider a thin uniform rod of length l and mass M , about an axis at one end.

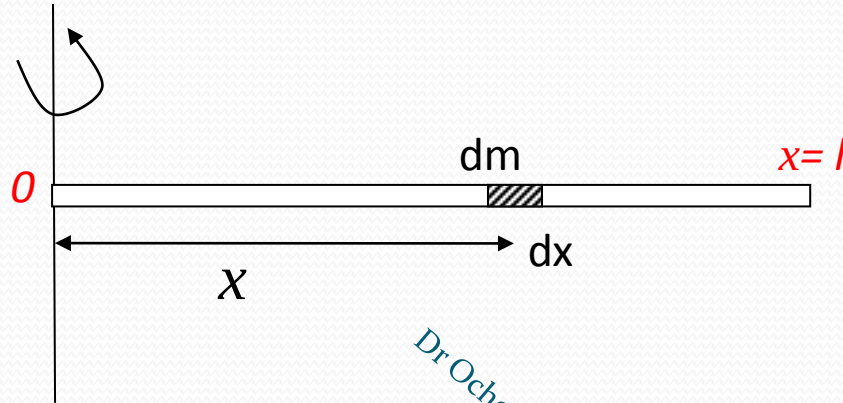
Assume it can be divided into suitable building blocks (point masses) of

- lengths dx and
- mass dm ,

Total mass = M , Length = l



Consider just **one element**, at some distance x , as in Fig.3.5, from the axis.



For the element (being a **point mass**),

Let its **moment of inertia** be **$dI = dm x^2$**

dm – can be written in terms of **mass** and **length** of rod

$$dm = \text{mass per unit length of rod} \times \text{the length of element } (dx)$$
$$\text{length of rod } (M/l)$$

i.e,

$$dm = \frac{M}{l} x dx$$

thus, $dl = \frac{M}{l} x^2 dx$, hence, $I = \int_0^l \frac{M}{l} x^2 dx$

The rod extends from $x=0$ to $x=l$ on the x-axis.

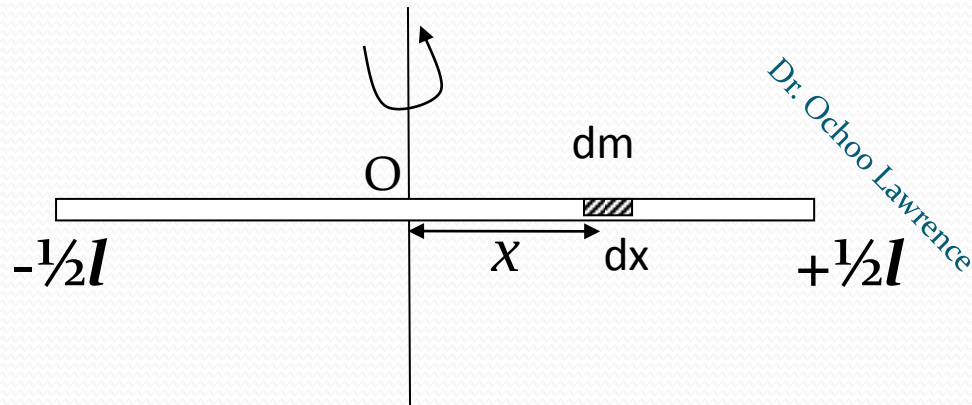
$$I = \int_0^l \frac{M}{l} x^2 dx = \left[\frac{M}{3l} x^3 \right]_0^l = \frac{M}{3l} [l^3 - 0]$$

thus, $I = \frac{1}{3} Ml^2$ (an axis at one end)

Case 2: (The axis at the centre)

- Consider the same procedure, when the axis is at the Centre of the rod.

Note: the ends are now at $x = -\frac{1}{2}l$ and $x = +\frac{1}{2}l$ on the x- axis



For the **point mass** element, let $dI = dm x^2$

Let

$$dm = \frac{M}{l} x dx$$

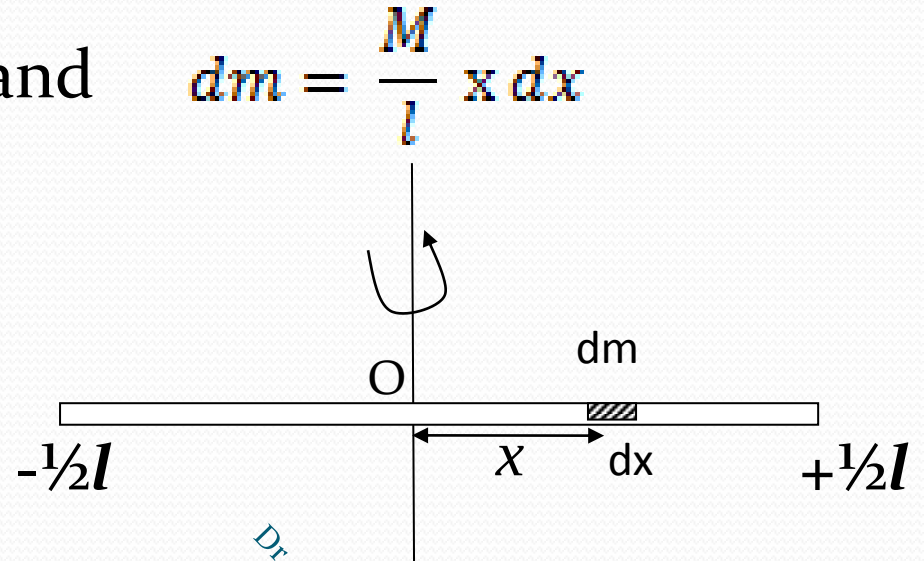
Therefore, from $dI = dm x^2$ and $dm = \frac{M}{l} x dx$

$$dI = \frac{M}{l} x^2 dx$$

Let, $I = 2 \int_0^{l/2} \frac{M}{l} x^2 dx$

$$= 2 \left[\frac{M}{3l} x^3 \right]_0^{l/2} = 2 \left[\frac{M}{3l} x \frac{l^3}{8} - 0 \right]$$

thus, $I = \frac{1}{12} M l^2$ (axis at the center)



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The difference confirms the following:

- Unlike mass, **the moment of inertia** of a body **is not one value**, it depends **on the location of the axis** of rotation.
- It depends on the **shape of the object** – different for the **rod** and a **point mass** about the same axis.

Dr. Ochoa-Lawrence

b). A thin Uniform Rectangular Sheet (**2-D**)



Dr. Ochoo Lawrence

- What would be the **suitable building blocks** for it?

Think of strips of **the thin uniform rods**, they can form a Rectangle faster than point masses would do, Fig. 3.5.

Fig. 3.5



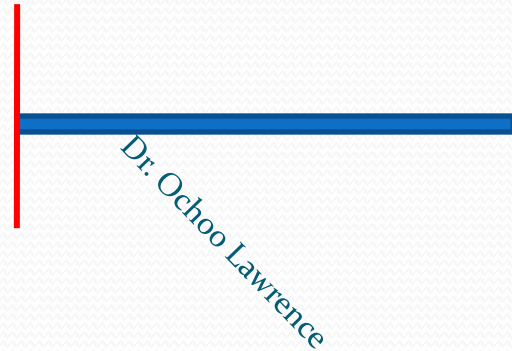
Moment of inertia, about any axis, is then the SUM of the **M.I for rods** about the **same location of axis**.

Note:

The obtained **forms** for the **moments of inertia** for a **thin rod** (about the axes), **are to guide** on how to write the formula for the **M.I** of rectangle.

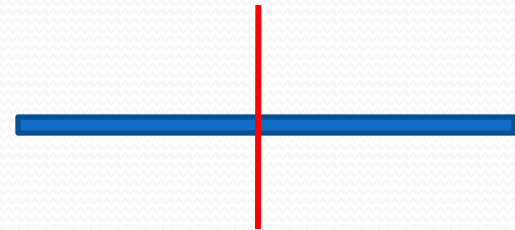
i.e for an **axis at one end**,
the form is

$$I = \frac{1}{3} Ml^2$$



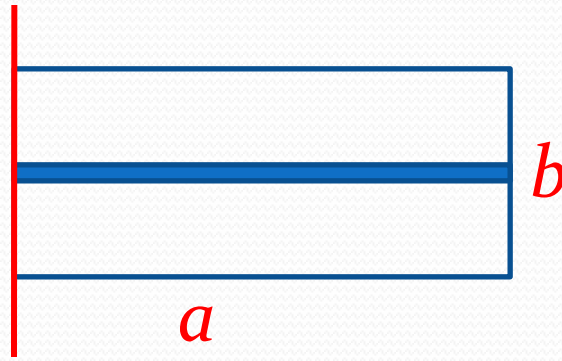
for **axis at the centre**,
the form is

$$I = \frac{1}{12} Ml^2$$



Case 1: (Axis at **one End**)

Consider a thin uniform rectangular sheet of mass **M**, length **a** and width **b** with an axis at the end as shown below.



Dr. Ochoo Lawrence

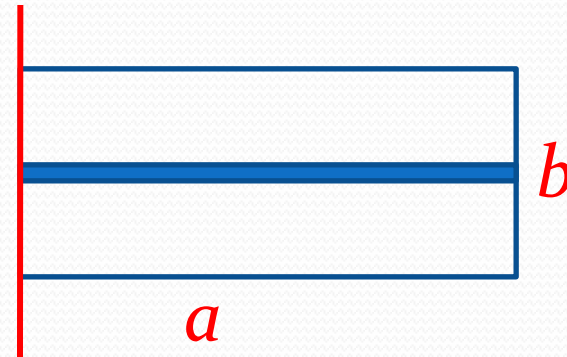
Starting with the moment of inertia of the **element (Rod)**

Whose **form**, about the shown axis, is $I = \frac{1}{3} Ml^2$

Thus :

If its mass be $= dm$,

length $l = a$,



then, let the M.I of the element be

$$dI = \frac{1}{3} dma^2$$

(one element)

For **2D** (**Area**), let $dm = \text{Mass/Area} \times \text{Area of element of object}$

- The Area of **the element** = length **x** width

- if the width be = dx
- then area of element = $a dx$

Hence, from $dI = \frac{1}{3} dma^2$

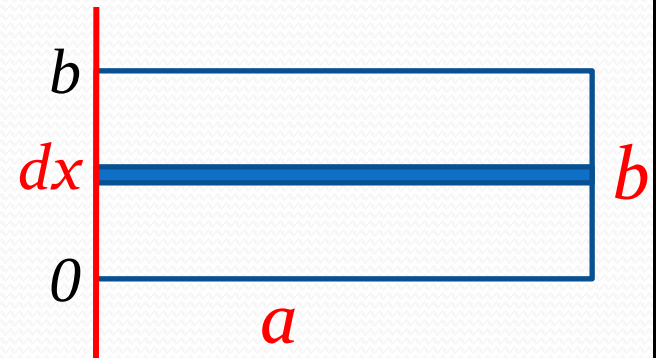
dm = Mass/Area $\times$ area of element of object

$$dm = \frac{M}{ab} \cdot a dx = \frac{M}{b} \cdot dx$$

Hence, $dI = \frac{1}{3} \frac{M}{b} \cdot a^2 dx$

thus, $I = \frac{1}{3} \frac{M}{b} \cdot a^2 \int_0^b dx = \frac{1}{3} \frac{M}{b} \cdot a^2 [x]_0^b$

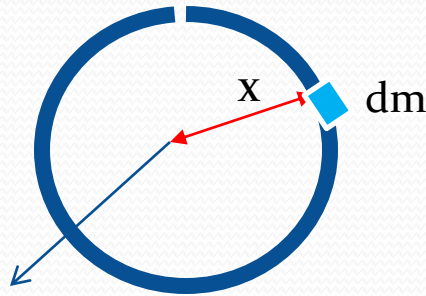
$$I = \frac{1}{3} Ma^2$$



c) Circular /Round shapes (**1D, 2D, 3D**)

For **circular** objects, the building units can be:

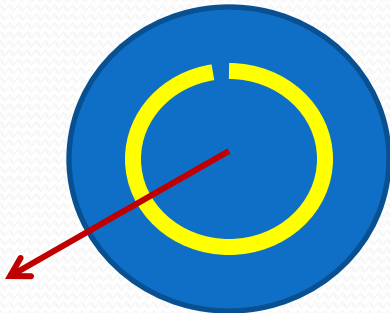
- A **hoop**/ring (1D) - Element (**point mass**)



$$dI = dm x^2$$

dm = Mass x length of element
per length

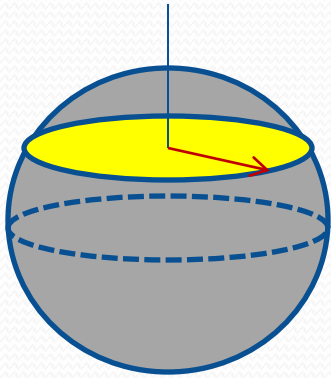
- A **disc** (2D) - Element (a **hoop**)



$$dI = dm x^2$$

dm = Mass x Area of element
per Area

- for a **cylinder** or **sphere** (3D) = Element (**a disc**)

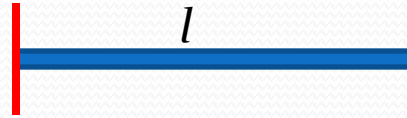


$$dI = \frac{1}{2} dm x^2$$

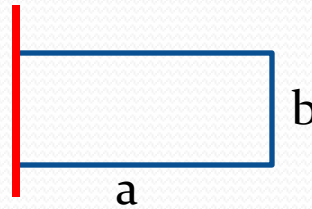
dm = Mass x vol. of element
per volume

When **integral calculus** is used, moments of inertia of such shapes would be as provided elsewhere:

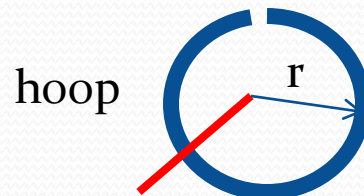
Generalization



$$I = \frac{1}{3} M l^2$$

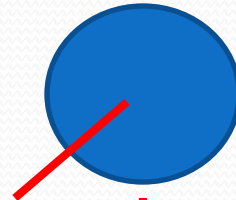


$$I = \frac{1}{3} M a^2$$



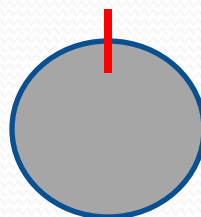
hoop

$$I = M r^2$$



Disc

$$I = \frac{1}{2} M r^2$$



sphere

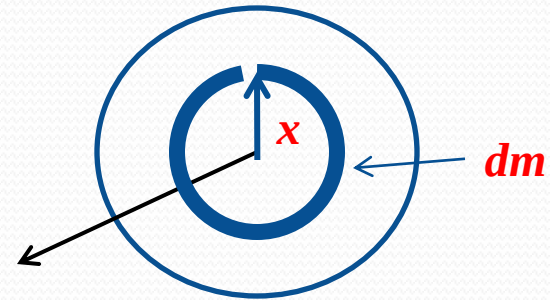
$$I = \frac{2}{5} M r^2$$

c) Thin uniform Disc -2D (axis through centre)

Consider a thin **uniform disc** of mass **M** and **radius r** about an axis of rotation **through the centre** of symmetry.

Its building unit is a **hoop** (rings)

- Let **element** hoop be:
- of mass **dm** and radius **x**



For a hoop M. of inertia

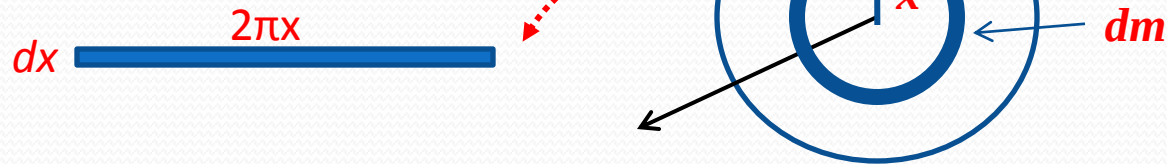
takes the form (**$I = Mr^2$**), hence, let **$dI = dm x^2$**

Therefore, from $dI = dm x^2$

For a **disc** (2-D).

dm = Mass/Area $\times$ area of element
of disc

Area of element = length $\times$ width



Hence, $dm = \frac{M}{\pi r^2} \cdot 2\pi x \cdot dx$

Therefore, $dI = \frac{M}{\pi r^2} \cdot 2\pi x \cdot x^2 dx$

$$I = \frac{2M}{r^2} \int_0^r x^3 dx = \frac{1}{2} Mr^2$$

Applications in Problem solving,
e.g, to determine **angular acceleration**

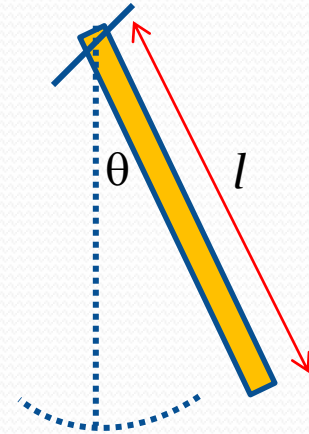
- Consider where we have real systems to determine some properties, like a **rigid body pendulum**, displaced by angle, as shown, and.

The Eq. for Rotational Motion is

$$\tau = I\alpha$$

So, you need

- **Torque** (τ) and **Moment of inertia** (I)



Recall;

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = |\mathbf{r}| |\mathbf{F}| \sin \theta$$

But what is $\mathbf{F} = ?$, what is $\mathbf{r} = ?$, and what is $\mathbf{I} = ?$

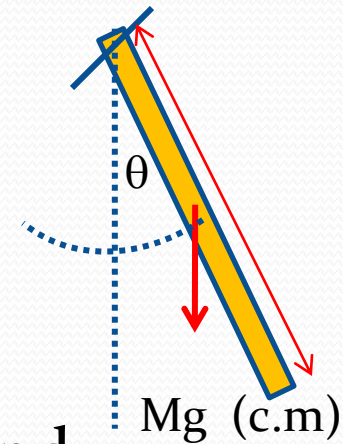
-Only known force is $\mathbf{Mg}$, at c.m

-so, $\mathbf{F} = \mathbf{Mg}$ and $r = \frac{1}{2}l$

hence, $\tau = \frac{1}{2}lMg \sin \theta$

Moment of inertia is about an axis at the end;

Let (if it is a rod) $I = \frac{1}{3}Ml^2$



Angular acceleration at that point $\alpha = \frac{\tau}{I} = \frac{3}{2} \frac{lMg \sin \theta}{Ml^2} = \frac{3}{2} \frac{g \sin \theta}{l}$

Parallel and Perpendicular Axes Theorems

There are times when **the locations** or **orientations** of axes **may shift**.

- General **quick** and **simple method** for determining **Moments of Inertia** about them may be necessary.
- Ideas which **relate** **parallel** and **perpendicular axes** have been realized **through mathematical** derivations

a). Parallel Axes Theorem

This relates axes of rotation which are parallel to one through the centre of mass of a body (May be regular or irregular).

It states that

the **moment of inertia** (I) about **any axis of rotation**, which may be **parallel to** an axis through the **centre of mass** at a distance **d** and whose moment of inertia is (I_o), is given as

$$I = I_o + Md^2 \quad \text{where } M \text{ – Mass of the body}$$

Examples

Consider determining moment of inertia (I) of a thin uniform rod of length l and mass M , about an axis at one end, given that M.I about axis at the c.m is $I_o = \frac{1}{12} Ml^2$

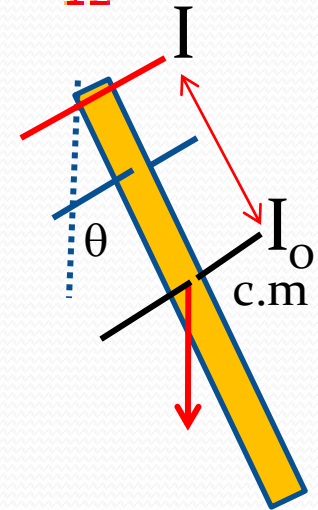
From $I = I_o + Md^2$

where M –mass, $d = \frac{1}{2}l$ and $I_o = \frac{1}{12} Ml^2$

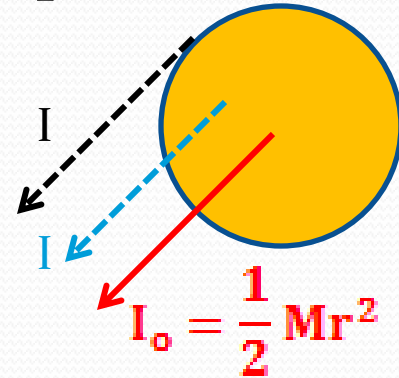
Thus,

$$I = \frac{1}{12} Ml^2 + M \left(\frac{l}{2} \right)^2 = \frac{1}{12} Ml^2 + \frac{1}{4} Ml^2$$

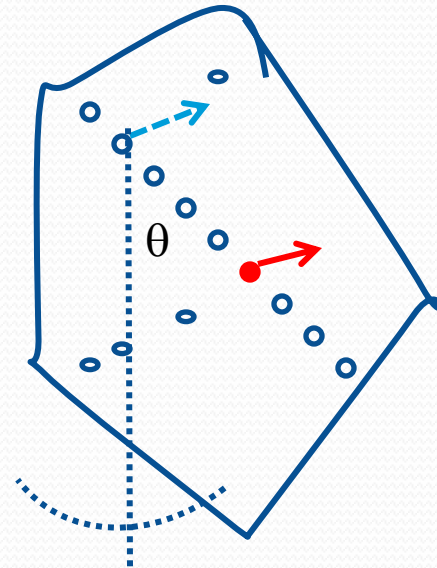
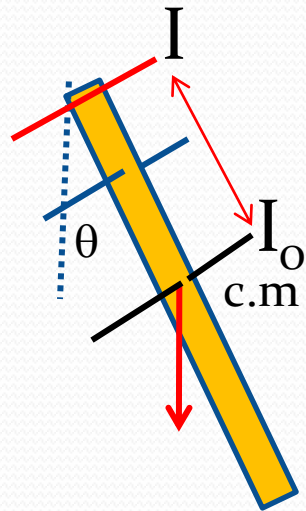
$$= \left(\frac{1}{12} + \frac{1}{4} \right) Ml^2 = \left(\frac{1+3}{12} \right) Ml^2 \quad I = \frac{1}{3} Ml^2$$



This theorem can apply to various shapes



Experimental setups in the Lab



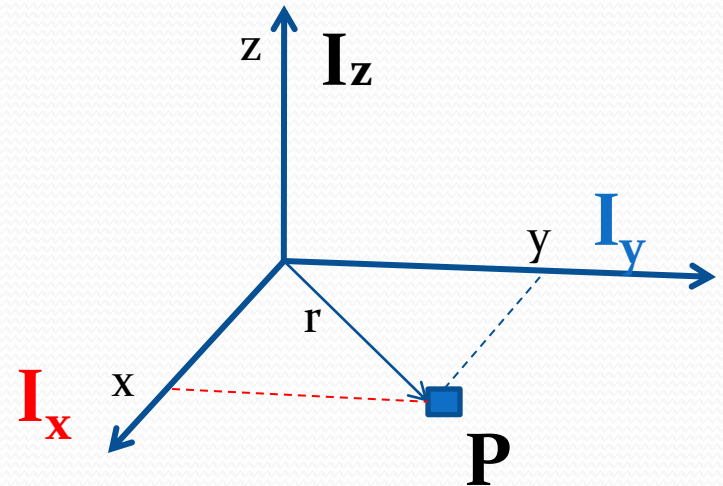
a). Perpendicular Axes Theorem

The theorem relates the Moments of inertia of a body about each of the axes as independent axes of rotation

- Its expression can be given in terms of the Moments of inertia about each of the three axes; x , y and z

- Choosing one of them as the axis of rotation,

- Say, if z -axis is the Axis of rotation of the point mass P



What are **the expressions** for the **moments of inertia** of P about x, y and z-axes?

From the general formula

$$I_i = m_i r_i^2$$

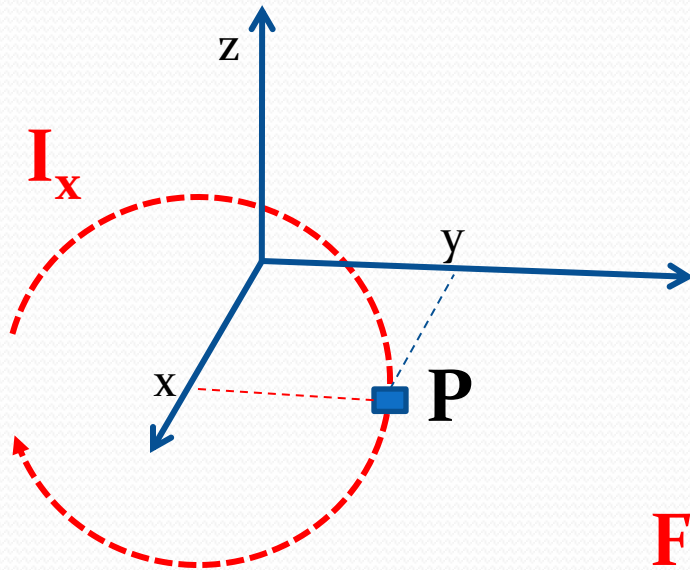


Fig. (a)

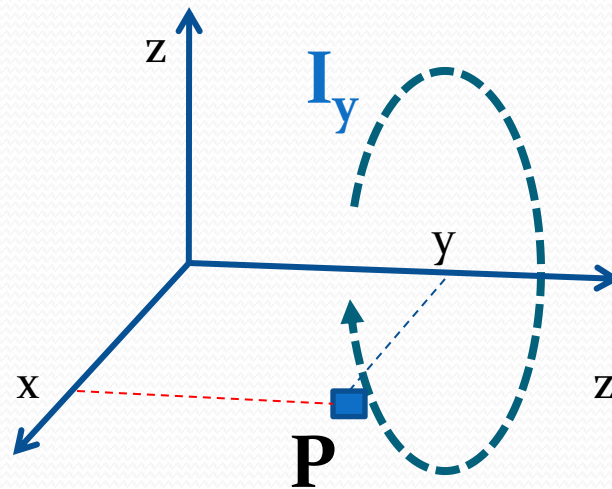


Fig. (b)

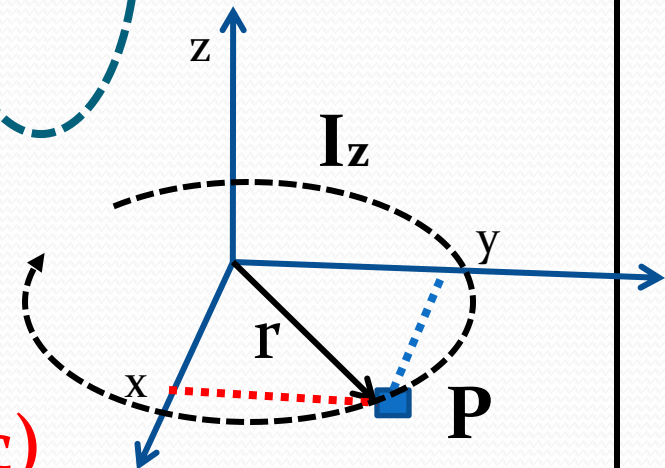


Fig. (c)

Moments of inertia, by $I_i = m_i r_i^2$

By the **main axis of Rotation** (**z-axis**)

(about z-axis) $I_z = m r^2$

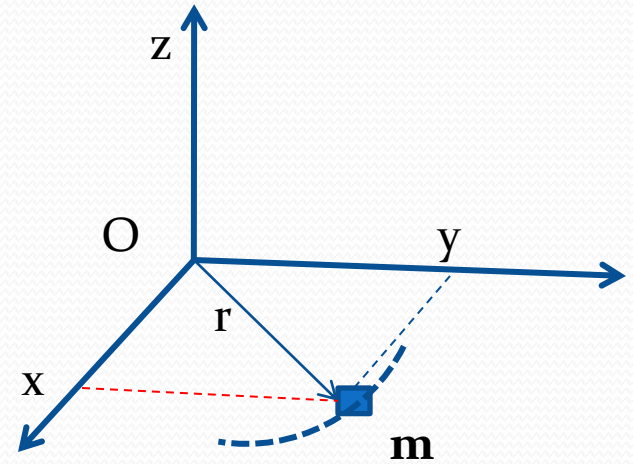
the (**r**) for z-axis is related to x and y
Axes by $r^2 = x^2 + y^2$

Hence, $I_z = m(y^2 + x^2) = m y^2 + m x^2$

By the moments of inertia

about **x-axis** $I_x = m y^2$

about **y-axis** $I_y = m x^2$ hence, $I_z = I_x + I_y$



It states that

the **moments of inertia of a body**, about three mutually perpendicular axes of rotation, like x, y and z axes, are related as:

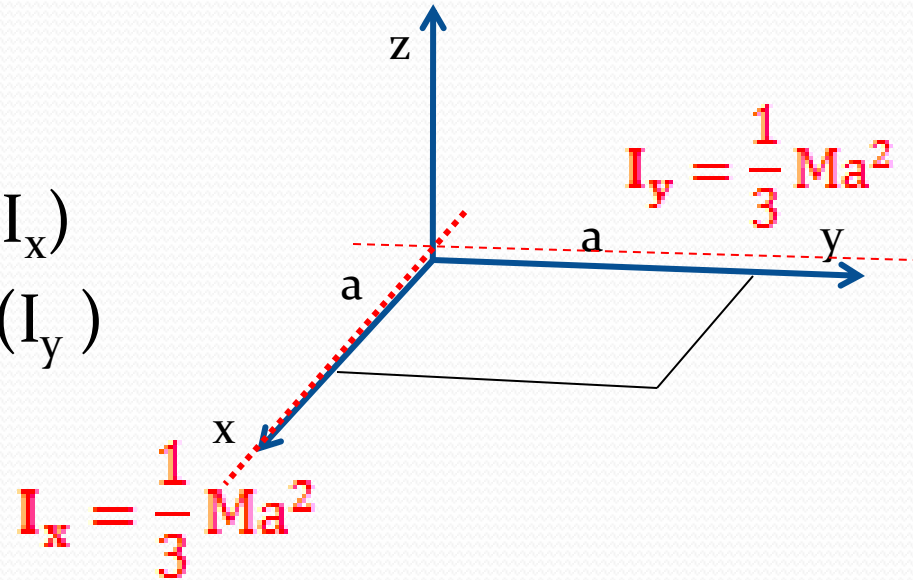
$$I_z = I_x + I_y$$

For z-axis as the axis of rotation

Example

If we consider a square sheet, lying in xy-planes and is to have angular motion about z-axis. What are its moments of inertia

- (i) About axis along x-axis (I_x)
- (ii) About axis along y-axis (I_y)
- (iii) About z-axis (I_z)



From $I_z = I_x + I_y$

$$I_z = \frac{1}{3}Ma^2 + \frac{1}{3}Ma^2$$

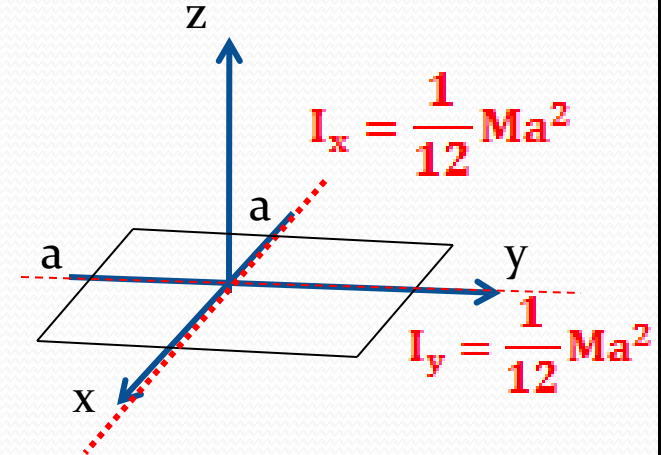
thus

$$I_z = \frac{2}{3}Ma^2$$

Example 2

If the axes were all through the centre of symmetry of the square (midpoint), as shown. What are its moments of inertia

- (i) About axis along x-axis (I_x)
- (ii) About axis along y-axis (I_y)
- (iii) About z-axis (I_z)

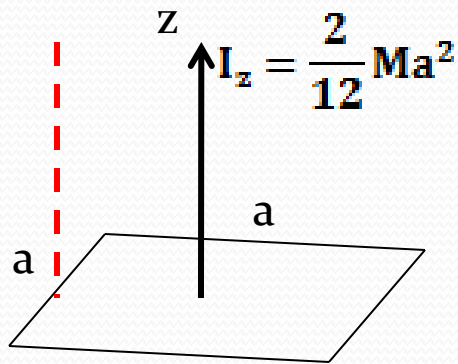


From $I_z = I_x + I_y$

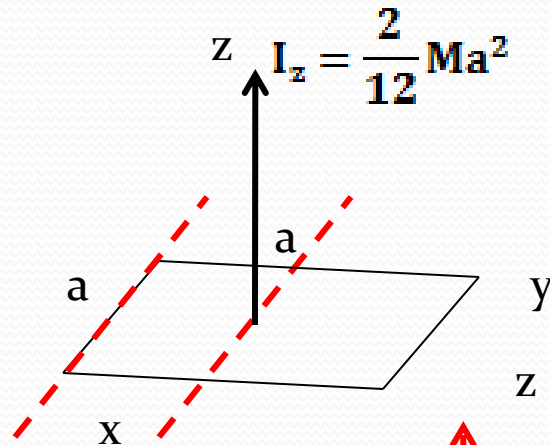
thus
$$I_z = \frac{2}{12}Ma^2$$

Combined Theorems (in problems)

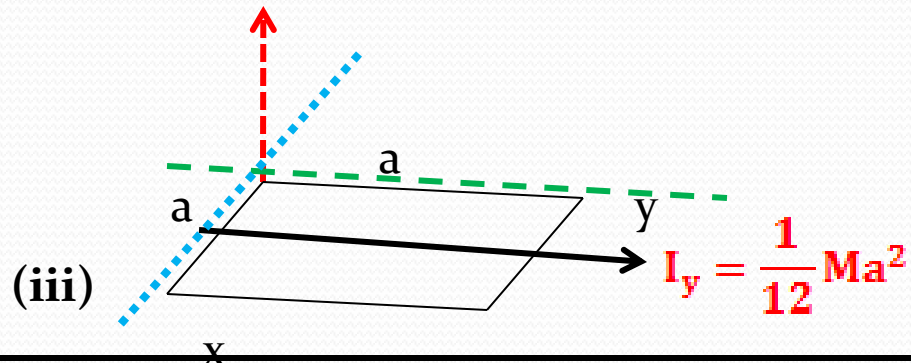
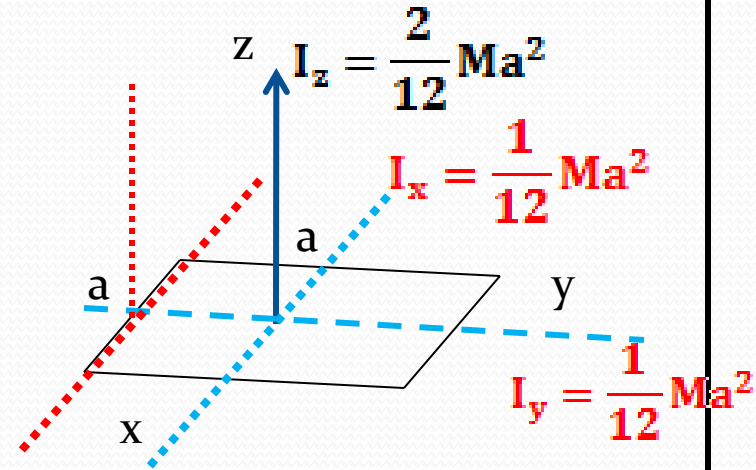
Sometimes, a problem may require combining two theorems . e.g, see Figs. Below. Think of moment of inertia about the red axes on each, starting with the black



(i)



(ii)



(iii)

3.3.6. Radius of gyration

For **every rigid body**:

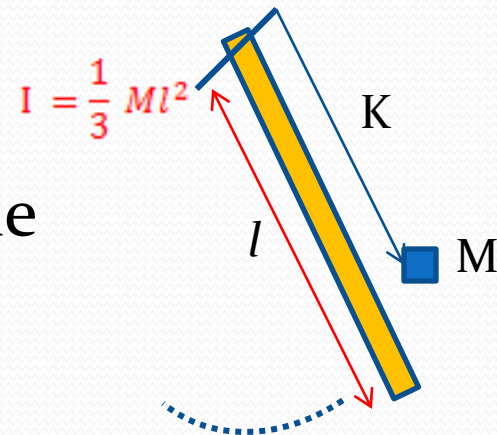
- there is **a distance K** from any axis of rotation, where its **whole mass M** can be Placed, **as point mass**, and its **moment of inertia (I)** is still the same about the axis.

The distance is called the **radius of gyration (K)**

Example

As shown, for an axis at one end of the Rod , for the radius of gyration K

Means $I = MK^2$ but also $I = \frac{1}{3} Ml^2$



Therefore, from $I = MK^2$

Means $K = \sqrt{\frac{I}{M}}$

And, for the rod, $I = \frac{1}{3} Ml^2$

Therefore, $K = \sqrt{\frac{\frac{1}{3} Ml^2}{M}}$

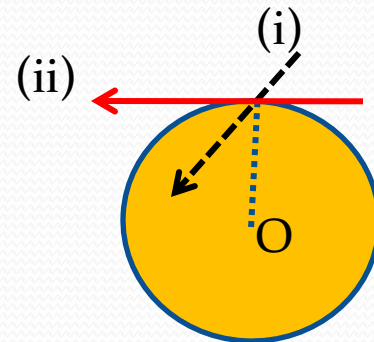
Hence, $K = \sqrt{\frac{l^2}{3}} = \frac{1}{\sqrt{3}} l$

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Assignment

Find **Moments of inertia** and **radius of gyration** of a thin uniform disc of radius **R** and mass **M**, for rotation about each of the two axis **through a point at the edge**, Fig. below:

- i). perpendicular to its plane **but parallel** to an axis through the centre, where Mom. of inertia at the centre is $I = \frac{1}{2}Mr^2$
- ii). Tangent to the edge



E-tivity 3.0.2

Provided in the Module

Do e-tivities Q1



LESSON 4

ANGULAR MOMENTUM AND ENERGY IN ROTATIONAL MOTION

LEARNING OUTCOMES

(**Provided elsewhere** – ensure you understand them)

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Introduction

Momentum and **energy** are two quantities which **define** a **state** of a body **because of velocity**.

- Any **moving object**, in **linear** or **angular motion**, has both **momentum** and **kinetic energy**
- A change in any, signifies a **change of state**.
- they may or may not be **conserved** /constant.

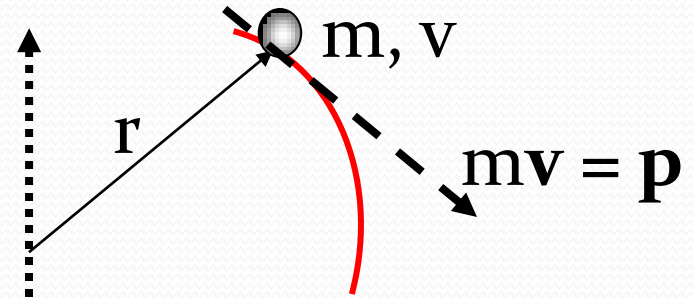
We need to obtain their Eqs., for **Rotational motion**, by **transformation** of Eqs. for linear motion of **a particle**

1). Angular Momentum in Rotational Motion

We now know that the **equations in linear motion** have **corresponding equations** in rotational motion, **obtainable by transformation**.

Thus, for **a point mass** of mass **m**, at position **r**, moving with a velocity **v**, ($p = mv$) as in Fig., angular momentum (**L** being vector) is by **cross product** as:

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = \mathbf{r} \times m\mathbf{v}$$



Example:

A particle of mass 2 units has a velocity $\mathbf{v} = 5\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$ at the point $\mathbf{r} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$. Find its angular momentum at that point.

$$\text{From } \mathbf{L} = \mathbf{r} \times m\mathbf{v} = m(\mathbf{r} \times \mathbf{v}) \quad \mathbf{L} = 2 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -3 \\ 5 & -3 & 6 \end{vmatrix}$$

$$= 2 \left\{ \mathbf{i} \begin{vmatrix} 2 & -3 \\ -3 & 6 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 1 & -3 \\ 5 & 6 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 1 & 2 \\ 5 & -3 \end{vmatrix} \right\}$$

$$= 2 \{3\mathbf{i} - 21\mathbf{j} - 13\mathbf{k}\} \quad \mathbf{L} = 6\mathbf{i} - 42\mathbf{j} - 26\mathbf{k}$$

From a **single particle** to a **rigid body** (many particles)

A rigid body is made of many particles

So, we sum up the expressions for individual particles

i.e from the expression $\mathbf{L} = \mathbf{r} \times m\mathbf{v} = \mathbf{r} \times \mathbf{p}$

For each particle, let, $\mathbf{L}_i = \mathbf{r}_i \times \mathbf{p}_i$

Therefore, let $\mathbf{L} = \sum \mathbf{r}_i \times \mathbf{p}_i = \sum \mathbf{r}_i \times m_i \mathbf{v}_i$

Let $\mathbf{L} = \sum m_i (\mathbf{r}_i \times \mathbf{v}_i)$

$\mathbf{v}_i$ –can be expressed in angular as $\mathbf{v}_i = \boldsymbol{\omega} \times \mathbf{r}_i$

Hence, let $\mathbf{L} = \sum m_i (\mathbf{r}_i \times (\boldsymbol{\omega} \times \mathbf{r}_i))$

From $\mathbf{L} = \sum m_i (\mathbf{r}_i \times (\boldsymbol{\omega}_i \times \mathbf{r}_i))$

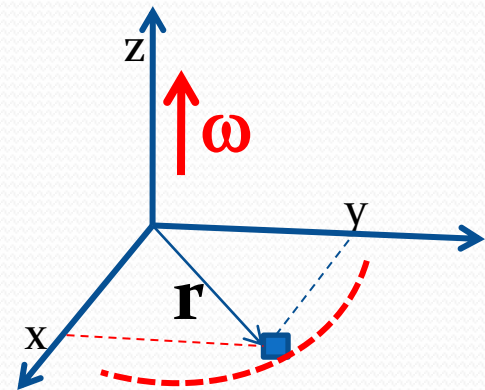
And identity $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B} (\mathbf{A} \cdot \mathbf{C}) - \mathbf{C} (\mathbf{A} \cdot \mathbf{B})$

$$\mathbf{L} = \sum m_i (\mathbf{r}_i \times (\boldsymbol{\omega}_i \times \mathbf{r}_i)) = \sum m_i \{ \boldsymbol{\omega}_i (\mathbf{r}_i \cdot \mathbf{r}_i) - \mathbf{r}_i (\mathbf{r}_i \cdot \boldsymbol{\omega}_i) \}$$

Dot product gives “Cos”.,

where , $\mathbf{r}_i \cdot \mathbf{r}_i = |\mathbf{r}_i| |\mathbf{r}_i| = r_i^2$

but $\mathbf{r}_i \perp \boldsymbol{\omega}_i$, hence, $\mathbf{r}_i \cdot \boldsymbol{\omega}_i = 0$



Therefore, $\mathbf{L} = \sum m_i r_i^2 \boldsymbol{\omega}_i$ but $m_i r_i^2 = I_i$

Thus , by the sum; $\mathbf{L} = \mathbf{I} \boldsymbol{\omega} \equiv \mathbf{p} = m\mathbf{v}$

Thus, there are **two expressions** for calculating **momentum**:

- If quantities are r , m and v ,

$$\mathbf{L} = \mathbf{r} \times m\mathbf{v} = \mathbf{r} \times \mathbf{p}$$

- If known quantities are I and ω

$$L = I \omega$$

ii). The law of Conservation of **Angular Momentum**

The condition **for change of state** starts from **Newton's second Law** of motion (LINEAR)

$$\mathbf{F} = m\mathbf{a}$$

Thus, for linear motion, **rate of change of momentum**

$$\mathbf{F} = \mathbf{ma} = m \frac{d\mathbf{v}}{dt} = \frac{d(m\mathbf{v})}{dt}$$

The **equivalent** in rotational motion – on **one particle**.

is $\tau_i = I_i \alpha_i$

Therefore, for a rigid body let $\tau = \sum \tau_i = \sum I_i \alpha_i$

But $\alpha_i = \frac{d\omega_i}{dt}$ (so as to **have angular velocity**)

$$\text{Therefore, } \tau = \sum I_i \frac{d\omega_i}{dt} = \sum \frac{d I_i \omega_i}{dt} = \frac{d \sum I_i \omega_i}{dt}$$

$$\text{Thus, } \tau = \frac{d I \omega}{dt} \equiv \mathbf{F} = \frac{d m \mathbf{v}}{dt}$$

Thus, by the two Newton's 2nd laws

$$\mathbf{F} = \frac{d \mathbf{mv}}{dt} \quad \equiv \quad \boldsymbol{\tau} = \frac{d \mathbf{I}\omega}{dt}$$

i.e **Net torque** is proportional to **rate of change** of **Ang. momentum**.

- Thus, if **net torque is zero**, there would be **no change** in Angular momentum (i.e $L = I \omega$ – **is conserved**).

i.e from $\boldsymbol{\tau} = \frac{d \mathbf{I}\omega}{dt} = I \frac{d\omega}{dt}$ if $\boldsymbol{\tau} = 0$

$$I \frac{d\omega}{dt} = 0 \quad \equiv \quad m \frac{dv}{dt} = 0$$

i.e there Should be **no change in velocity**

Therefore,
the **principle of conservation of angular momentum**
would require that *the Torque* $\tau = 0$

To state that:

- If there is **no net external torque** acting on a body in rotational motion, its **angular momentum** remains constant (**conserved**)

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In **Rotational Motion**, if $I\omega$ is to remain constant

i.e if $I\omega = k$ (constant) like $p = mv = k$

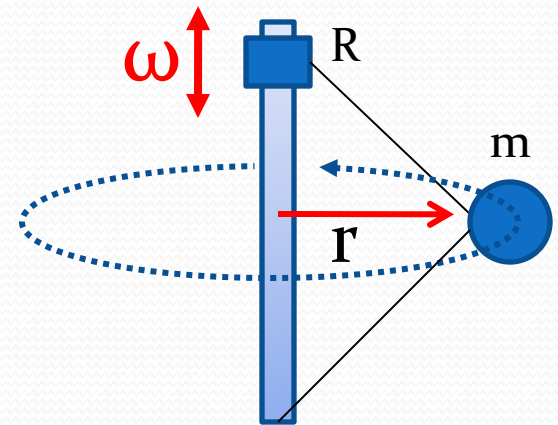
then, in case ω or I changes then the other should also Change/adjust to compensate for the change

- What does that mean?

From $I\omega = k \Rightarrow I \propto \frac{k}{\omega}$ (inverse)

Moment of Int.(I) $I = mr^2$

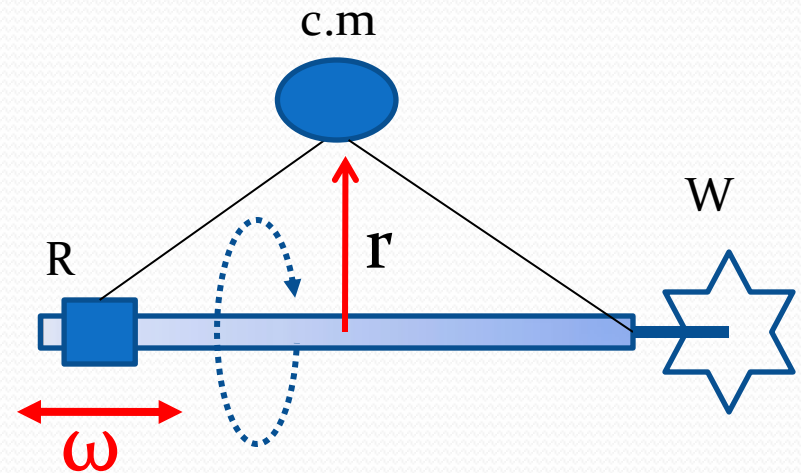
$$mr^2 \propto \frac{k}{\omega}$$



Innovative Application ideas

$$I \propto \frac{k}{\omega}$$

$$mr^2 \propto \frac{k}{\omega}$$



Increased stability, **lowering c.m** - Mercedes Benz
Speed governor – **fuel cutout**

2). Kinetic Energy in Rotational Motion

- Kinetic energy is a scalar quantity.
- If the kinetic energy K.E of a particle in Linear Motion is (T),

For a single particle $T_i = \frac{1}{2} m_i v_i^2$

Therefore, for all particles, in a rigid body we sum up as;

Total K.E
$$T = \sum T_i = \sum \frac{1}{2} m_i v_i^2$$

Then by transformation of $\mathbf{v} = \omega \mathbf{r}$,

$$T = \sum \frac{1}{2} m_i \omega_i^2 r_i^2 \quad \text{let} \quad T = \frac{1}{2} \sum m_i r_i^2 \omega_i^2 = \frac{1}{2} \sum I_i \omega_i^2$$

$$T = \frac{1}{2} I \omega^2 \equiv K.E = \frac{1}{2} M \mathbf{v}^2$$

Example (Rotational K.E)

A uniform thin rod of length 0.8m rotates about an axis through its centre, with an angular velocity of 5 rad/sec. If its mass is 4 kg, find its kinetic energy of rotation.

From, rotational K.E $T = \frac{1}{2} I \omega^2$

Where, $I = \frac{1}{12} M l^2$ for the axis at the centre

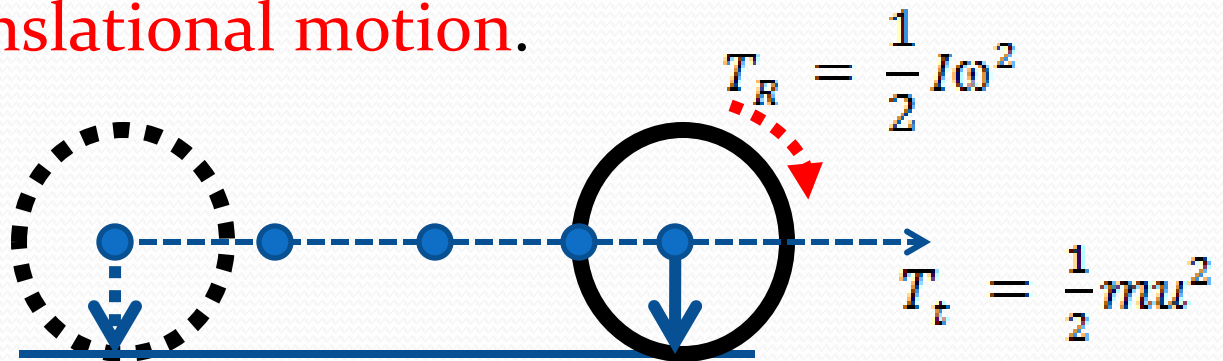
Therefore, $T = \frac{1}{2} \left(\frac{1}{12} M l^2 \right) \omega^2$



$$T = \frac{1}{2} \left(\frac{1}{12} \times 4 \times 0.8^2 \right) \times 5^2 = 2.67 \text{ J}$$

Translational Rotational Motion of a Rigid body

A body in rotational motion may **not be at a fixed point**, it may also be in **translational motion**.



- Thus, the C.Mass is **shifted in linear displacement** as it also rotates.
- Note; **every pattern** of motion has **associated kinetic energy**. [units = Joules (J)]

Therefore, kinetic energy is **in two modes**;

- Translational (**linear**) K.E $T_t = \frac{1}{2}mu^2$

- Rotational K.E. $T_R = \frac{1}{2}I\omega^2$

Total K.E (T) $T = \frac{1}{2}mu^2 + \frac{1}{2}I\omega^2$

E-tivity 4.0

Provided in the Module

Ensure :

- you read and understand what to be done.
- You do the stated e-tivities.



LESSON 5

WORK –ENERGY RELATION AND POWER

LEARNING OUTCOMES

(**Provided elsewhere** – ensure you understand them)

Introduction (**Work –Energy/Power Relation**)

In physics;

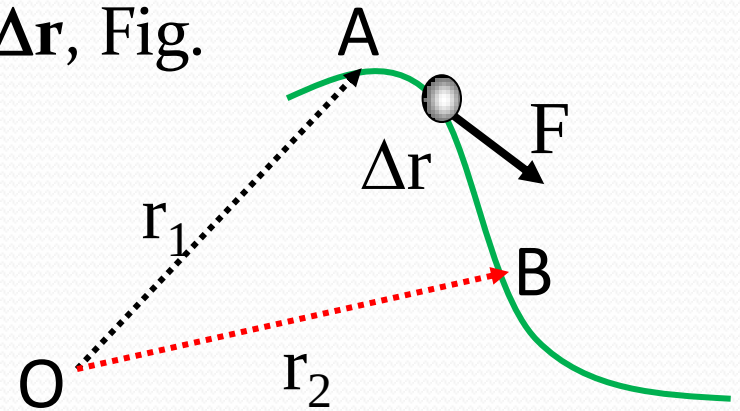
- Work is **energy** a body can **gain or lose** when acted upon by some force.

When an **external force /force field** acts on a body it can store /add **energy to the body** (as **Kinetic** or **potential** energy)

- Thus, being energy, work is **a Scalar quantity**, and
- it is calculated as a **product of force** and **displacement**, as

$$\Delta W = \mathbf{F} \bullet \Delta \mathbf{r}$$

Consider a **point mass** moved by force F , from point A to B, along the path **AB**, a displacement Δr , Fig.



The work done can then be expressed as:

i). $\Delta W = \mathbf{F} \cdot \Delta \mathbf{r}$ (Apply when F – **Constant**)

ii). $\partial W = \mathbf{F} \cdot \partial \mathbf{r}$ (when F - **varies with the coordinates**
lim $\Delta r \rightarrow 0$ (x,y,z), i.e F is a function of x,y, z.

Example

Force given by $\mathbf{F} = 2\mathbf{i} - \mathbf{j} + 7\mathbf{k}$ N acts on a body over a displacement given by $\mathbf{r} = 3\mathbf{i} + 2\mathbf{j} + \mathbf{k}$

. Determine the work done over that displacement.

From $W = \mathbf{F} \cdot \Delta \mathbf{r}$

$$W = (2\mathbf{i} - \mathbf{j} + 7\mathbf{k}) \cdot (3\mathbf{i} + 2\mathbf{j} + \mathbf{k})$$

$$= 6 - 2 + 7$$

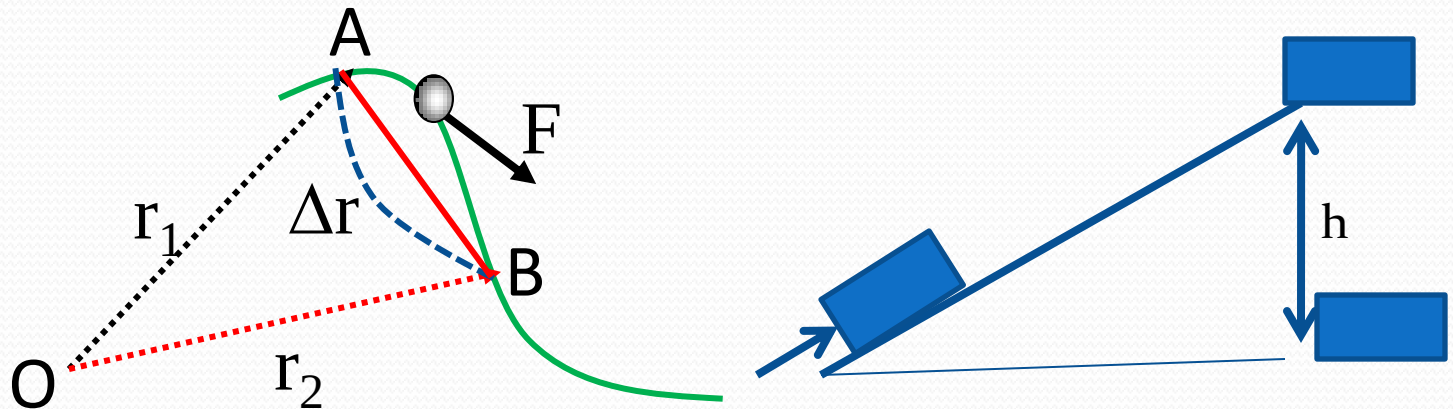
$$W = 11 \text{ Nm or } 11 \text{ J}$$

In terms of **position vectors**,

Let AB ; $\Delta \mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$

Hence, $\Delta W = \mathbf{F} \bullet (\mathbf{r}_2 - \mathbf{r}_1) = \mathbf{F} \bullet \mathbf{r}_2 - \mathbf{F} \bullet \mathbf{r}_1$

- Which shows that work done between two points is equal to the **Energy difference** between the two points.
- This is suggesting that **the route taken** may not matter.



There are cases where the **work done**

- may **not depend** on the **route taken** by a moving body.
- may depend on the route taken

For the case in which **force is dependent on the position** $\mathbf{F}(\mathbf{x}, \mathbf{y}, \mathbf{z})$, the appropriate expression for work becomes;

$$dW = \mathbf{F} \cdot d\mathbf{r}$$

Leading to $W = \int \mathbf{F} \cdot d\mathbf{r}$

- where $d\mathbf{r} = i \, dx + j \, dy + k \, dz$

- And $\mathbf{F} = F_{xyz}i + F_{xyz}j + F_{xyz}k$

- Therefore, we would have

$$W = \int \mathbf{F} \cdot d\mathbf{r}$$

$$W = \int (F_{xyz}\mathbf{i} + F_{xyz}\mathbf{j} + F_{xyz}\mathbf{k}) \cdot (\mathbf{i} dx + \mathbf{j} dy + \mathbf{k} dz)$$

By the Dot product, the vector components eliminate,
Hence,

$$W = \int F_{xyz} dx + \int F_{xyz} dy + \int F_{xyz} dz$$

Example

A force given as $F = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$ moves a body from the point $(1, -2, 1)$ to $(3, 1, 4)$. Find the work done.

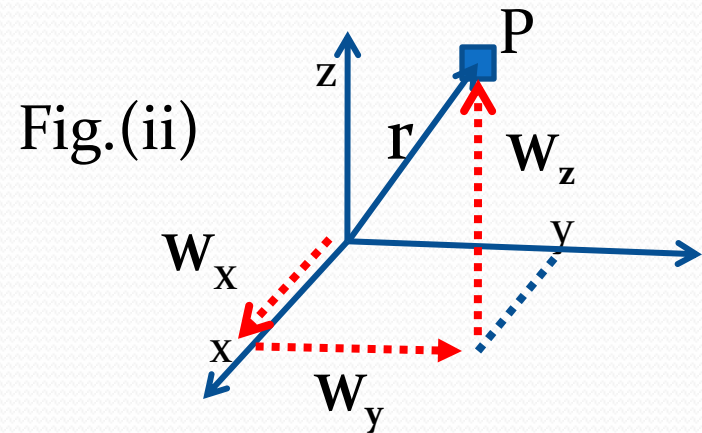
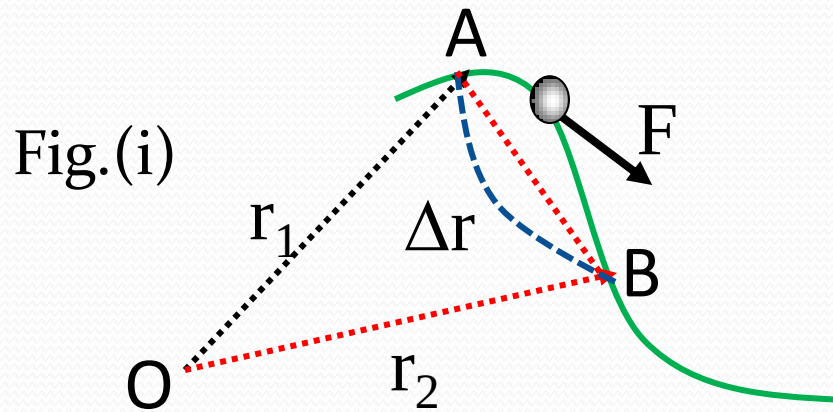
Here the force **varies** with **position**, therefore; we can have:

$$W_x = \int F_{xyz} dx = \int (2xy + z^3) dx = x^2y + xz^3 + C_x$$

$$W_y = \int F_{xyz} dy = \int x^2 dy = x^2y + C_y$$

$$W_z = \int F_{xyz} dz = \int 3xz^2 dz = xz^3 + C_z$$

Where work(**Energy**) is **independent** of the route/path but on the **end-points**, meaning values will be the **same** along different routes.



As a result, **some terms** are **the same** as observed in Eqs.

$$W_x = x^2y + \mathbf{xz^3} + C_x$$

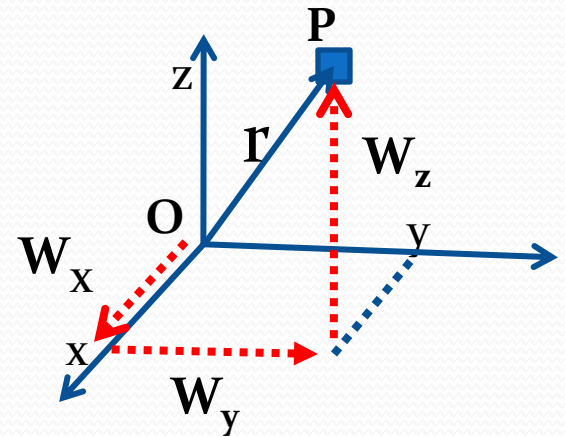
$$W_y = x^2y + C_y$$

$$W_z = \mathbf{xz^3} + C_z$$

So, we need to **pick** and **sum** unique/similar **terms** only once, to **obtain work done** between **the end-points**, (OP) as

$$W_{OP} = W_x + W_y + W_z$$

i.e
$$W = x^2y + xz^3 + C$$



We can therefore **calculate work** from the limits given from the point $(1, -2, 1)$ to $(3, 1, 4)$

i.e from

$$F = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$$

$$r_1(1, -2, 1) \text{ to } r_2(3, 1, 4).$$

Whose work expression is $W = x^2y + xz^3 + C$

Therefore, let
$$W = [x^2y + xz^3 + C]_{1,-2,1}^{3,1,4}$$

$$= [(9 \times 1) + (3 \times 4^3)] - [1 \times (-2) + 1]$$

$$W = 201 + 1 = \mathbf{202 \text{ Nm}}$$

A Case where the work done by **force/force field** is **independent of the path**, the force is described as **conservative force /force field**

5.3. Conservative forces/force field

Definition:

A **conservative force field** is a force field in which the **work done** on a body moved from one point to the other is **independent of the route**.

Mathematically, conservative force fields (**F**) **obey the Condition**.

$$\nabla \times \mathbf{F} = \mathbf{0}$$

read as $\text{Curl } \mathbf{F} = 0$

Where, $\nabla = i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z}$ is a **vector operator** del

With the vector components of the force, the expression

$$\nabla \times \mathbf{F}$$

is evaluated by matrix method, as follows;

$$\nabla \times \mathbf{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$$

$$= i \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_y & F_z \end{vmatrix} - j \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ F_x & F_z \end{vmatrix} + k \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ F_x & F_y \end{vmatrix}$$

$$= i \left(\frac{\partial}{\partial y} F_z - \frac{\partial}{\partial z} F_y \right) - j \left(\frac{\partial}{\partial x} F_z - \frac{\partial}{\partial z} F_x \right) + k \left(\frac{\partial}{\partial x} F_y - \frac{\partial}{\partial y} F_x \right)$$

Example

A certain force field is described by the expression .

$$\mathbf{F} = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$$

Determine if it is a conservative force field or not.

For a conservative force field, $\nabla \times \mathbf{F} = \mathbf{0}$

• Where $\nabla = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$

• let
$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix}$$

Therefore, from

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^3 & x^2 & 3xz^2 \end{vmatrix}$$

$$= \mathbf{i} \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & 3xz^2 \end{vmatrix} - \mathbf{j} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ 2xy + z^3 & 3xz^2 \end{vmatrix} + \mathbf{k} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ 2xy + z^3 & x^2 \end{vmatrix}$$

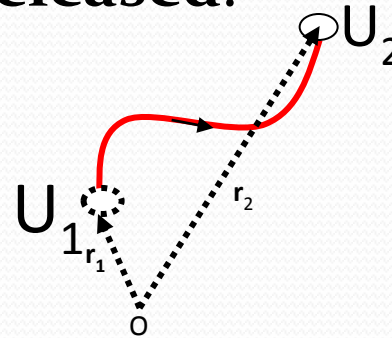
$$= \left(\frac{\partial}{\partial y} 3xz^2 - \frac{\partial}{\partial z} x^2 \right) \mathbf{i} - \left(\frac{\partial}{\partial x} 3xz^2 - \frac{\partial}{\partial z} (2xy + z^3) \right) \mathbf{j} + \left(\frac{\partial}{\partial x} x^2 - \frac{\partial}{\partial y} (2xy + z^3) \right) \mathbf{k}$$

$$= (0 - 0) \mathbf{i} - (3z^2 - 3z^2) \mathbf{j} + (2x - 2x) \mathbf{k} = \mathbf{0}$$

Hence, $\nabla \times \mathbf{F} = \mathbf{0}$ a conservative force field

5.2. Work and Potential energy relation

When a body is **moved in force field**, from one point to the other, **potential energy** is **stored** or **released**.



- The **stored** or **released** energy is equal to the work done.
- It is equal to the **difference in potential** energy between the two points. U_2 and U_1

The **Work-potential Energy** relation is

$$\Delta W = - \Delta U \quad \text{OR} \quad \partial W = - \partial U \quad (-ve - \text{each point is from } \infty)$$

By this relation,

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$$\Delta W = \mathbf{F} \cdot \Delta \mathbf{r}$$

Then, **work**, **force** and **potential energy** is related to force also by:

$$\Delta W = - \Delta U = \mathbf{F} \cdot \Delta \mathbf{r}$$

OR

$$\partial W = - \partial U = \mathbf{F} \cdot \partial \mathbf{r}$$

Thus, we have the following relations to use

$$\partial W = \mathbf{F} \cdot \partial \mathbf{r} \quad (\text{i})$$

$$\partial W = - \partial U \quad (\text{ii})$$

$$- \partial U = \mathbf{F} \cdot \partial \mathbf{r} \quad (\text{iii})$$

From the expressions

$$\partial W = - \partial U = \mathbf{F} \cdot \partial \mathbf{r}$$

Various important combinations can be formed for determining any of the quantities.

e.g, we have seen for

$$1) \quad \partial W = \mathbf{F} \cdot \partial \mathbf{r} \quad \partial \mathbf{r} = \mathbf{i} \partial x + \mathbf{j} \partial y + \mathbf{k} \partial z$$

$$\mathbf{F} = \frac{\partial W}{\partial \mathbf{r}} \quad \text{where} \quad \frac{\partial}{\partial \mathbf{r}} = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$$

$$2). - \partial U = \mathbf{F} \cdot \partial \mathbf{r}$$

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- $U = - \int_{r_1}^{r_2} \mathbf{F} \cdot \partial \mathbf{r}$

- $\mathbf{F} = - \frac{\partial U}{\partial \mathbf{r}}$

Therefore,

- let $\mathbf{i} \frac{\partial W}{\partial x} = \mathbf{i} \frac{\partial}{\partial x} (x^2 y + xz^3 + C) = (2xy + z^3) \mathbf{i}$

$$\mathbf{j} \frac{\partial W}{\partial y} = \mathbf{j} \frac{\partial}{\partial y} (x^2 y + xz^3 + C) = x^2 \mathbf{j}$$

$$\mathbf{k} \frac{\partial W}{\partial z} = \mathbf{k} \frac{\partial}{\partial z} (x^2 y + xz^3 + C) = 3xz^2 \mathbf{k}$$

Hence, $\mathbf{F} = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$

The same as the original case for determining the work done.

$$\mathbf{F} = (2xy + z^3)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}.$$

Some **force fields are not conservative**, meaning the work done would vary with the route taken,

For such cases, the **work done cannot be calculated** from the **potentials of the end points**.

$$\Delta W \neq U_2 - U_1$$

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Exercise

If all the terms from the x, y and z components in previous example were to be added we'd have

$W = 2x^2y + 2xz^3 + C$. Determine expression for the force field.

Solution: - Note; this is a **work-force relation**

Therefore, from $\mathbf{F} = \frac{\partial W}{\partial \mathbf{r}}$, but $\frac{\partial}{\partial \mathbf{r}} = \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z}$

• Hence;

$$\mathbf{F} = \frac{\partial W}{\partial \mathbf{r}} = \left(\mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y} + \mathbf{k} \frac{\partial}{\partial z} \right) W$$

Which can be written as;

$$\mathbf{F} = \mathbf{i} \frac{\partial W}{\partial x} + \mathbf{j} \frac{\partial W}{\partial y} + \mathbf{k} \frac{\partial W}{\partial z}$$

From

$$\mathbf{F} = \left(i \frac{\partial W}{\partial x} + j \frac{\partial W}{\partial y} + k \frac{\partial W}{\partial z} \right)$$

$$= i \frac{\partial}{\partial x} (2x^2y + 2xz^3 + C) + j \frac{\partial}{\partial y} (2x^2y + 2xz^3 + C) + k \frac{\partial}{\partial z} (2x^2y + 2xz^3 + C)$$

$$\mathbf{F} = (4xy + 2z^3)\mathbf{i} + 2x^2\mathbf{j} + 6xz^2\mathbf{k}$$

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Exercise

Find the expression for **potential U** associated with the force field given as

$$\mathbf{F} = (y^2 - 2xyz^3)\mathbf{i} + (3 + 2xy - x^2z^3)\mathbf{j} + (6z^3 - 3x^2yz^2)$$

From $\partial W = -\partial U = \mathbf{F} \cdot \partial \mathbf{r}$

$$U = - \int_{r_1}^{r_2} \mathbf{F} \cdot \partial \mathbf{r} \quad \text{but} \quad d\mathbf{r} = \mathbf{i} dx + \mathbf{j} dy + \mathbf{k} dz$$

Therefore,

$$U = - \int_{r_1}^{r_2} \{(y^2 - 2xyz^3)\mathbf{i} + (3 + 2xy - x^2z^3)\mathbf{j} + (6z^3 - 3x^2yz^2)\mathbf{k}\} \cdot (\mathbf{i} \partial x + \mathbf{j} \partial y + \mathbf{k} \partial z)$$

$$U = - \left(\int (y^2 - 2xyz^3) \partial x + \int (3 + 2xy - x^2z^3) \partial y + \int (6z^3 - 3x^2yz^2) \partial z \right)$$

Express as

$$U_x = - \int y^2 - 2xyz^3 dx = -xy^2 + x^2yz^3 + C_x$$

$$U_y = - \int 3 + 2xy - x^2z^3 dy = -3y - xy^2 + x^2yz^3 + C_y$$

$$U_z = - \int 6z^3 - 3x^2yz^2 dz = -\frac{3}{2}z^4 + x^2yz^3 + C_z$$

Since, $-U = W$ (Pick **unique terms**)

Hence, Let

$$U = - \left(xy^2 + 3y - x^2yz^3 + \frac{3}{2}z^4 + C \right)$$

Exercise

Verify that given **the potential** (scalar) due to conservative force field as:

$$U = - \left(xy^2 + 3y - x^2yz^3 + \frac{3}{2}z^4 + C \right)$$

The force field is a vector given by;

$$\mathbf{F} = (y^2 - 2xyz^3)\mathbf{i} + (3 + 2xy - x^2z^3)\mathbf{j} + (6z^3 - 3x^2yz^2)\mathbf{k}$$

Hint: From $\partial W = - \partial U = \mathbf{F} \cdot \partial \mathbf{r}$

E-tivity 5.0

Provided in the Module

Ensure :

- you read and understand what to be done.
- You do the stated e-tivities.

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LESSON 6

OSCILLATORY MOTION

LEARNING OUTCOMES

(**Provided elsewhere** – ensure you understand them)

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Introduction

Oscillatory motion is a repeating **to-and –fro** movement of a body between two points, about an equilibrium point.

- can be **repeating after equal intervals** of time (**Periodic**)
- The **periodic** oscillatory motion is also described as **harmonic motion**
- Our everyday life is supported by **natural** and **technological simple** and **complex** oscillatory motion

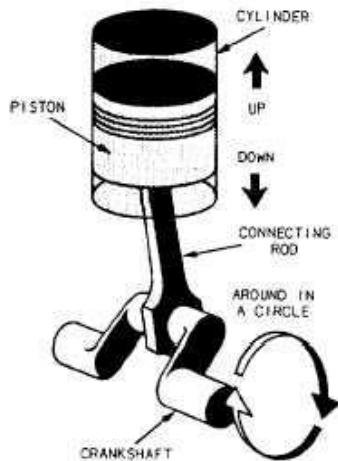
Examples are:

- human heart, lungs, wall clocks, vehicle engines, water pumps among many more.

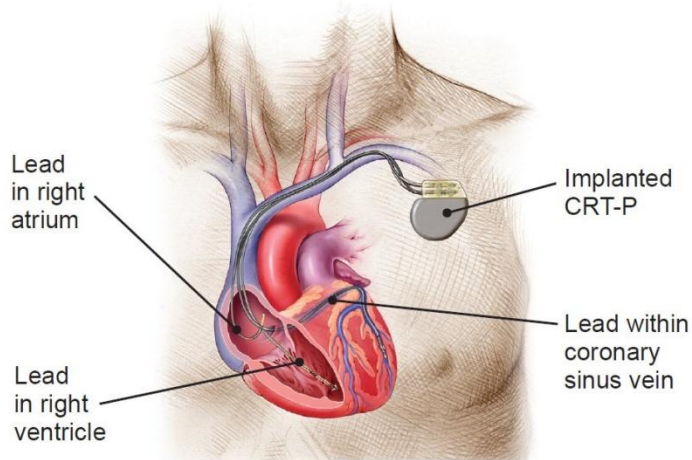
By studying **oscillatory motion** and the **laws of physics** which explain them:

- we can **understand** and **design** systems whose operations depend on oscillatory motion.

e.g vehicles, heart pacer



Heart Pacer- senses when heart is slow or fast and sends a signal to the heart to change the pace



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Two **important property** of oscillatory motion are:

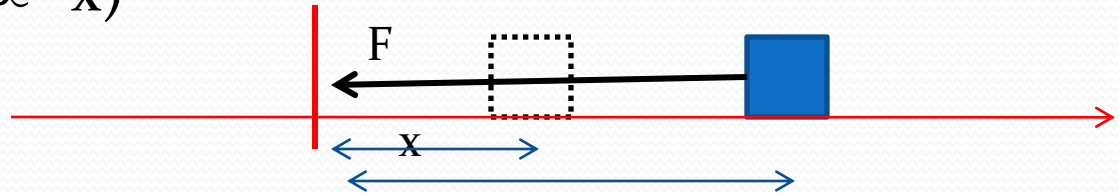
- requires a restoring force to maintain it
- It involves energy conversion from one form to the other in a system

Oscillatory motions **whose displacement** with **time** can be described by **simple mathematical functions** like “**Sin**” are described as **simple harmonic motion (SHM)**.

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For **simple harmonic motion (SHM)**, the **restoring force (F)** meets two conditions:

- it is proportional to the displacement (x) of a body from the equilibrium ($F \propto x$) - **increases** with displacement
- it is always directed in the direction opposite that of displacement ($F \propto -x$)



Definition:

Simple Harmonic Motion - motion whose **net restoring force (F)** is **proportional** to the **displacement (x)** and always **directed opposite** the direction of **displacement** as $(F \propto -x)$.

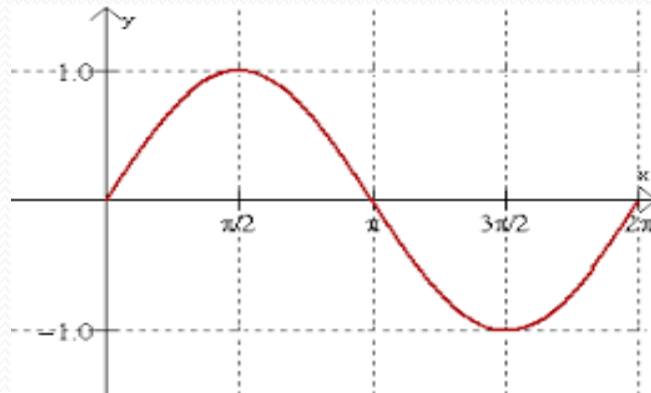
The concept of **net force** on a body is expressed in the **Newton's second law** $F = ma$

Therefore, even in SHM, the same law applies and can be expressed in a **differential Eq. form**

$$F = ma = m \frac{d^2 x}{dt^2} \quad \text{where} \quad a = \frac{d^2 x}{dt^2}$$

Where x – is the displacement which can be in **Sin / Cos**;
signifying oscillation;

e.g $x = A \sin (wt + \phi)$



Thus, from

$$F = ma = m \frac{d^2 x}{dt^2}$$

if the **displacement** (**x**) of a body of mass **m** is known, then the following can be determined:

- its **acceleration** (**a**)
- the **net restoring force** **F** on it
- the relation between **F** and displacement **x**.

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Example

Consider the case of a body of mass m in oscillation whose displacement is described by $x = A \sin \omega t$.

Determine if its motion is SHM (i.e if $F \propto -x$).

we determine acceleration, F , then the relation between F and x (if it appears as: $F \propto -x$)

From the Differential Equation: $F = ma = m \frac{d^2 x}{dt^2}$

Since x is known, a and F can be obtained as

$$a = \frac{d^2 x}{dt^2} = \frac{d^2}{dt^2} A \sin \omega t = -\omega^2 A \sin \omega t$$

Therefore,

$$F = m \frac{d^2 x}{dt^2} = -m \omega^2 A \sin \omega t$$

Since, $A \sin \omega t = x$ then $F = -m \omega^2 x$

thus, $F \propto -x$ hence, SHM

Thus, any oscillatory motion whose displacement is in terms of **Sin** OR **Cos** is SHM.

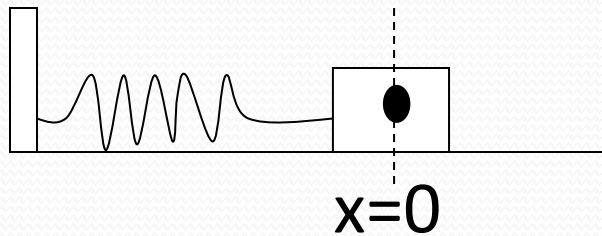
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The **characteristics** of oscillatory motion that are of interest are properties like:

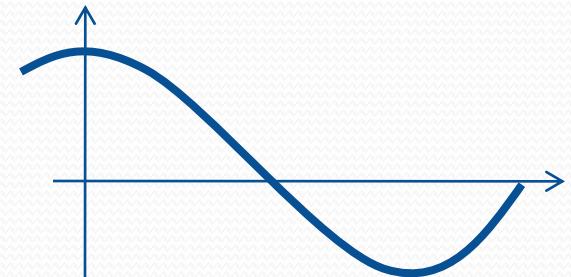
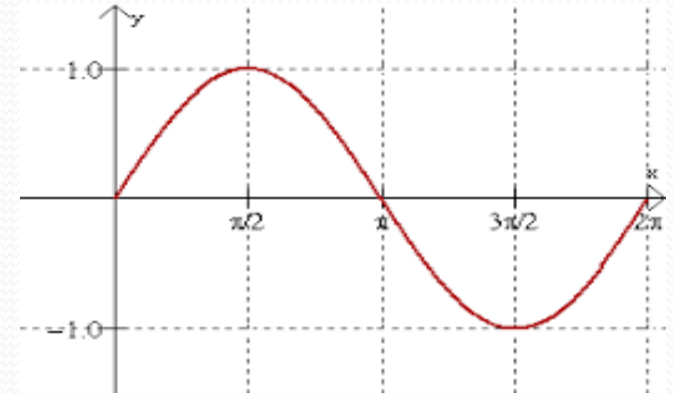
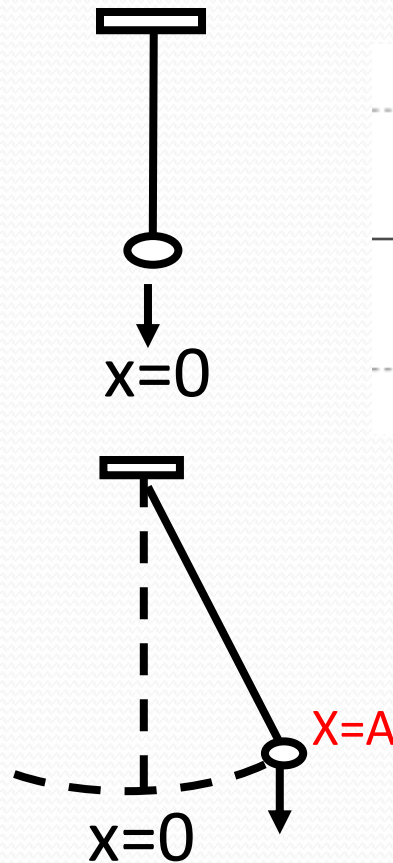
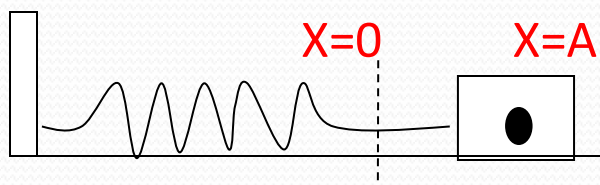
- Amplitude – maximum displacement
- Frequency – number of oscillations per second
- Period – related to frequency, the duration it takes to make one complete oscillation.
- Velocity /acceleration at various points
- Energy management/conservation of a system- interchange between potential and kinetic energies.

In general: **for all systems in SHM**, the displacement obeys either **Sin or Cos** function (depending on displacement at $t = 0$)

e.g $x = A \sin(\omega t \pm \theta)$



$x = A \cos(\omega t \pm \theta)$



Considering one case

e.g $x = A \sin (\omega t \pm \theta)$

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For a real system whose restoring net force appears in the form $F = -kx$, meaning SHM,

Where k –is based on the physical quantities of the system.

A number of quantities can be determined at any position x

And from the differential equation for SHM motion;

$$F = m \frac{d^2 x}{dt^2} \quad \text{which means} \quad -kx = m \frac{d^2 x}{dt^2}$$

Thus, for **SHM**, the **Differential Eq.** can be solved by taking

$$x = A \sin(\omega t \pm \theta) \quad \text{or} \quad x = A \cos(\omega t \pm \theta),$$

i.e, from $F = -kx = m \frac{d^2 x}{dt^2}$

$$a = \frac{d^2 x}{dt^2} = -\omega^2 x$$

Hence, $-kx = -m\omega^2 x$ **and** $k = m\omega^2$
therefore

$$\omega = \sqrt{\frac{k}{m}}$$

Then, **frequency** ($\omega = 2\pi f$) and **period** ($T = 1/f$) can be determined **in terms of the quantities** represented by k

i). A Simple Pendulum

Consider a pendulum of length l swinging in a vertical Plane. If motion is to the right of the equilibrium point.

- At angle θ (a displacement x), the forces acting along the line of displacement give a **net restoring force** as:

Net F = Pulling - Opposing

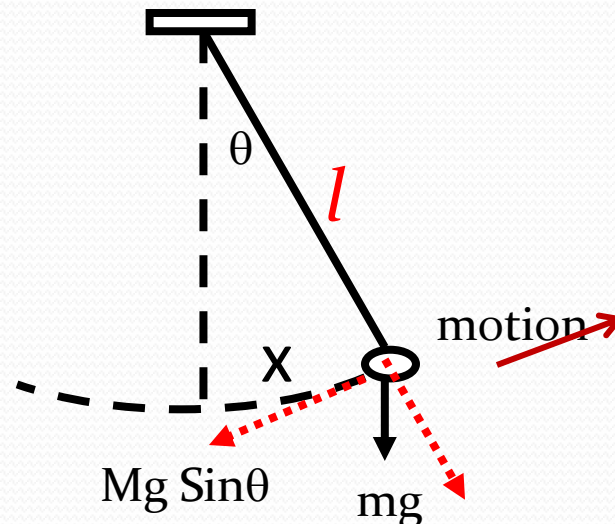
$$F = 0 - mg \sin \theta$$

For small angles,

$$\sin \theta \approx \theta$$

Hence, $F = -mg\theta$

$$F \propto -\theta \quad (\text{SHM})$$



Or, from the diagram, angle θ is related to displacement (x) by (from $s = r\theta$);

$$x = l\theta \quad \text{thus} \quad \theta = \frac{x}{l}$$

Hence, from $F = -mg\theta$

$$F = -mg\frac{x}{l}$$

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Thus, $F \propto -x$ (SHM)

Hence, we can determine frequency and period of the pendulum motion, by applying differential equation

- i.e, for systems in SHM, the differential equation, given by

$$F = m \frac{d^2x}{dt^2}$$

leads to **$F = -m\omega^2x$** .

Thus, for a simple pendulum we set

$$F = -mg \sin \theta = -m\omega^2x$$

Thus, for small angles **$-mg \theta = -m\omega^2x$**

And since $\theta = \frac{x}{l}$

Let **$-mg \frac{x}{l} = -m\omega^2x$** , therefore, **$\frac{g}{l} = \omega^2$**

Therefore, from

$$\frac{g}{l} = \omega^2 = (2\pi f)^2$$

the **frequency** and **periodic time** of a pendulum, would be

$$f = \frac{1}{2\pi} \sqrt{\frac{g}{l}}$$

And from $T = \frac{1}{f}$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

Example

Given a simple pendulum of length **0.9m**, calculate its periodic time T and frequency f . Take $g = 10\text{m/s}^2$.

Solution,

i). Periodic time; From $T = 2\pi \sqrt{\frac{l}{g}}$

$$\begin{aligned} T &= 2 \times 3.14 \times \sqrt{\frac{0.9}{10}} \\ &= 1.9\text{sec} \end{aligned}$$

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ii). Mass on a Spring

Consider another case of SHM; a **block of mass** m attached to a spring of spring **constant** k and moving on a frictionless horizontal surface, Fig.

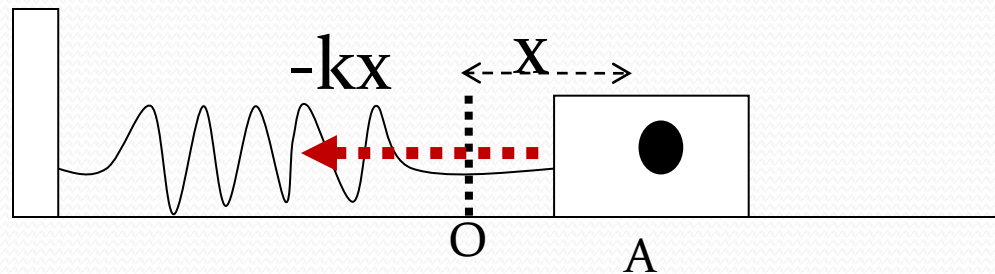
- At the **displacement** x **of the centre of mass** to the right of O , **at point A**. The acting force is tension in the spring ($F_s = kx$), towards the left

Therefore,

Net force

$$F = 0 - kx$$

$$F = -kx \quad (\text{SHM})$$



Therefore, by the differential expression for SHM, Net force

$$F = m \frac{d^2 x}{dt^2} = - m \omega^2 x$$

Hence, $-kx = - m \omega^2 x$

So, $\omega^2 = \frac{k}{m}$

Hence, $f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$ and $T = 2\pi \sqrt{\frac{m}{k}}$

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Other quantities in SHM, like **velocity, Amplitude, Kinetic energy, acceleration** , can be determined from the general displacement Eq. in SHM. $x = A \cos (\omega t \pm \theta)$,

e.g Determining **Velocity** and **Acceleration** at various points

from; $v = \frac{dx}{dt}$ then $v = \frac{d}{dt} A \cos (\omega t \pm \theta) = -\omega A \sin (\omega t \pm \theta)$

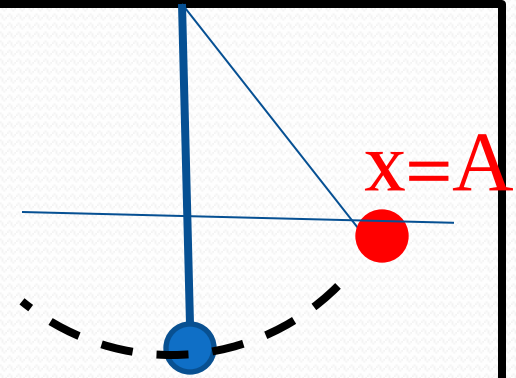
$$v = -\omega A \sin (\omega t \pm \theta)$$

- Depending on the location of **displacement** x , by the angle

- At $x=0$; hence, from $x = A \cos(\omega t \pm \theta)$,

$$0 = A \cos(\omega t \pm \theta), \text{ thus, } (\omega t \pm \theta) = 90$$

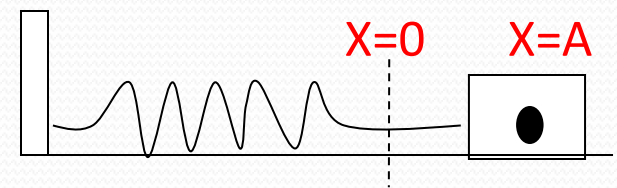
$$\text{Hence, } v = -\omega A \sin(90) = -A\omega \text{ (Max)} \quad x=0$$



- At (max. displacement) $x = A$,
means $A = A \cos(\omega t \pm \theta)$, therefore, $(\omega t \pm \theta) = 0$

$$\text{Hence, } v = -\omega A \sin(\omega t \pm \theta)$$

$$v = -\omega A \sin(0) = 0$$



For **acceleration** (Rate of change of velocity)

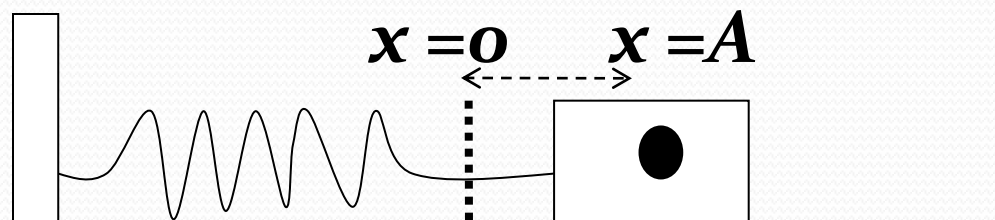
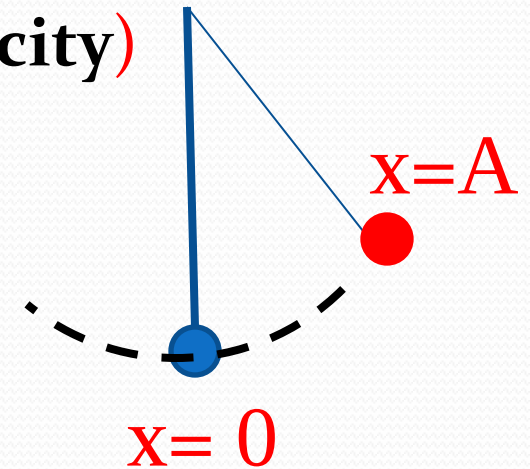
$$a = \frac{d^2x}{dt^2}$$

- For $x = A \cos(\omega t \pm \theta)$,

$$a = -\omega^2 A \cos(\omega t \pm \theta)$$

$$a = -\omega^2 x \quad \text{i.e rate of change of vel. depends on } x$$

- Thus, acceleration at any **displacement** x can be obtained.



Kinetic and Potential Energy (in SHM)

In SHM, the oscillator converts energy from one form to the other (K.E –to-P.E – to- K.E) but the total energy remains Constant.

By total energy $E = K.E(T) + P.E (U)$

From $K.E = \frac{1}{2}mv^2$ and $x = A\cos(\omega t \pm \theta)$,

therefore $v = \frac{dx}{dt} = -\omega A \sin(\omega t \pm \theta)$

Hence $K.E(T) = \frac{1}{2}M\omega^2 A^2 \sin^2(\omega t \pm \theta)$

For P.E (U),

We recall, $-dU = Fdx$

$$\text{Let } U = P.E = \int -F dx$$

But for SHM, $F = -kx$

$$\text{Hence, } P.E = \int -(-kx) dx \quad \text{hence, } U = P.E = \frac{1}{2}kx^2$$

But, since $x = A \cos(\omega t \pm \theta)$,

$$\text{Therefore, } P.E = \frac{1}{2}kA^2 \cos^2(\omega t \pm \theta)$$

Hence, the Total energy in SHM is

$$E = \frac{1}{2} m \omega^2 A^2 \sin^2(\omega t \pm \theta) + \frac{1}{2} k A^2 \cos^2(\omega t \pm \theta)$$

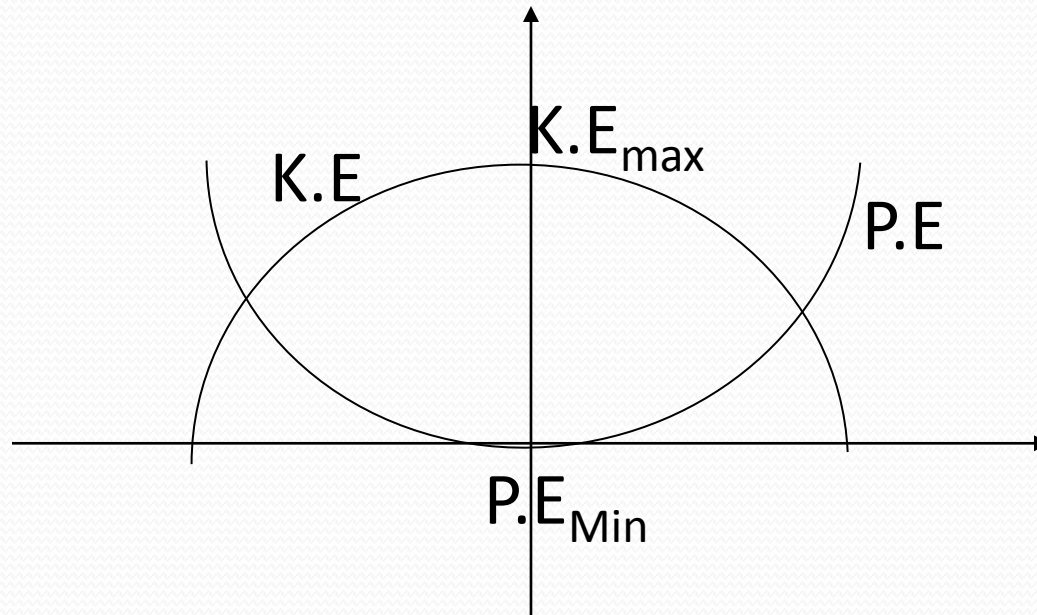
But $m \omega^2 = k$ (constant)

$$\text{Hence, } E = \frac{1}{2} k A^2 \sin^2(\omega t \pm \theta) + \frac{1}{2} k A^2 \cos^2(\omega t \pm \theta)$$

$$E = \frac{1}{2} k A^2 [\sin^2(\omega t \pm \theta) + \cos^2(\omega t \pm \theta)]$$

$$\text{Thus, Total energy } E = \frac{1}{2} k A^2 \quad (\text{Constant and } \propto A^2)$$

The variations in K.E and P.E can be represented as



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Forces on a single particle at various points

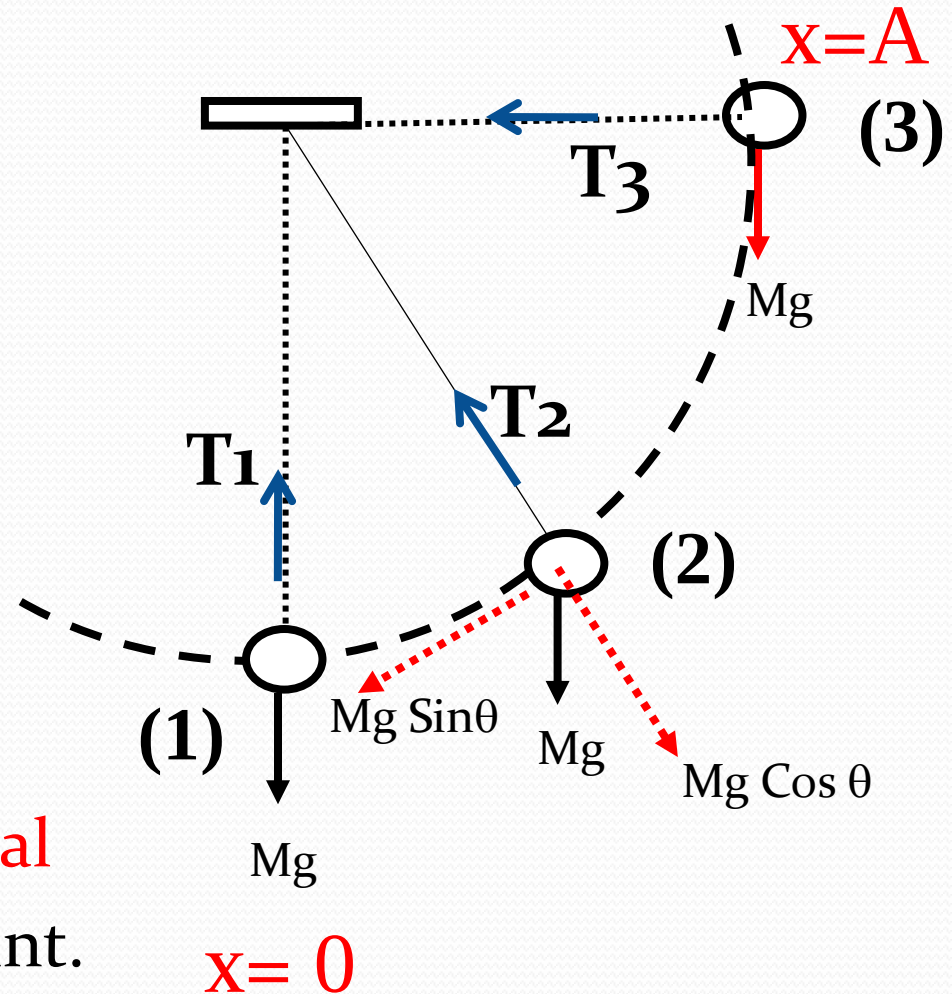
Where is **Tangential acceleration**

- Max?
- Zero?

Why?

$$a = -\omega^2 x$$

There would also be **radial Acceleration** at every point.



A Compound Pendulum (rotation in rigid bodies)

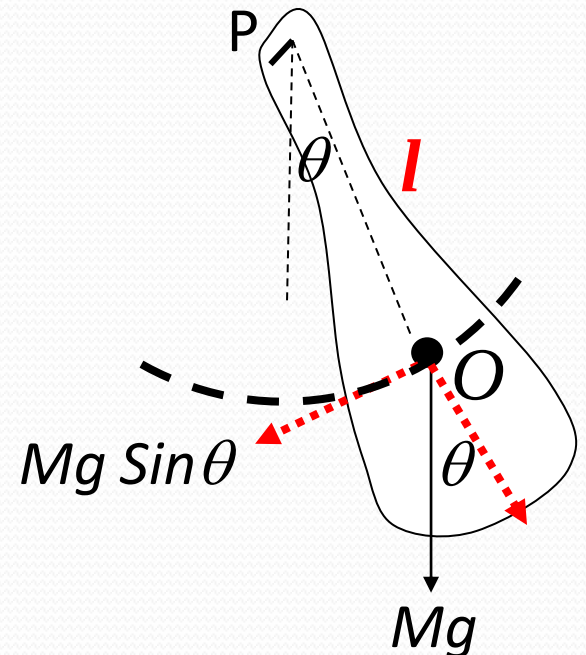
- We now consider condition for SHM of an oscillating rigid body, called compound pendulum.
- In this case we consider torque instead of force .

Consider pendulum mass M ,
C.M at O , a distance l from axis P
displaced by angle θ

by the acting forces, we get Net torque

$$\text{Net Torque } \tau = 0 - Mgl \sin \theta$$

$$\tau \approx -Mgl\theta \quad (SHM)$$



Then, **by Equation of Motion** (Newton's 2nd Law) we can determine quantities like frequency and periodic time.

- i.e, from $\tau = I\alpha$

Let $-Mgl\theta = I\alpha$

But linear acceleration $a = \frac{d^2x}{dt^2} = -\omega^2x$

- And $a = r\alpha = l\alpha$

let $\alpha = a/l = -\frac{\omega^2x}{l}$ and from $x \approx l\theta$

- let $\alpha = -\frac{\omega^2l\theta}{l} = -\omega^2\theta$

- Therefore; $-Mgl\theta = -I\omega^2\theta$

Thus,

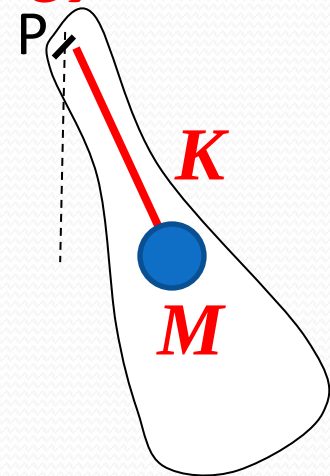
$$\omega^2 = \frac{Mgl}{I} \quad \text{and since} \quad \omega = 2\pi f$$

Frequency (f); $f = \frac{1}{2\pi} \sqrt{\frac{Mgl}{I}}$ and $T = 2\pi \sqrt{\frac{I}{Mgl}}$

But I – can be expressed in terms of **radius of gyration K**,
Recall, by definition (K)

“Distance from axis where whole mass
can be placed without altering (I)

i.e $I = MK^2$



Mg

Thus, by substituting $I = MK^2$ into $T = 2\pi \sqrt{\frac{I}{Mgl}}$

We get $T = 2\pi \sqrt{\frac{K^2}{gl}}$ Note: K and l may be different

Compared to T for simple pendulum of length l_o ; $T = 2\pi \sqrt{\frac{l_o}{g}}$

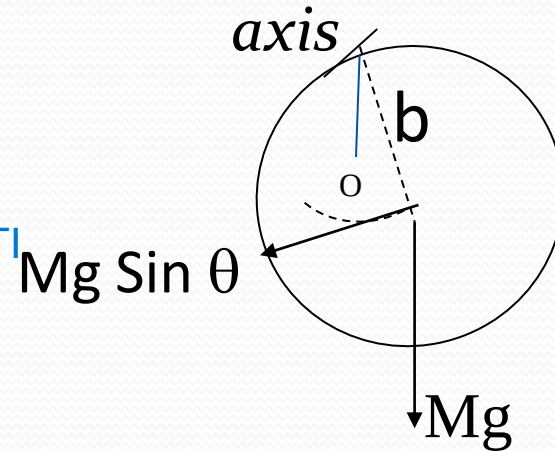
Sometimes, one may think of substituting a compound pendulum with a simple pendulum which would still have same f or T . What should be the length l_o ?

If a simple pendulum has to have same T , $l_o = \frac{K^2}{l}$

Exercise

A uniform disc of mass M and radius b is set to oscillate in a vertical plane, about an axis at its edge and perpendicular to its plane, as shown in the diagram. (moment of inertia about an axis through C.M of a disc and perpendicular to the plane is, $I_o = \frac{1}{2} MR^2$)

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Determine : i). expression for its periodic time as a compound pendulum.

ii). the length of its equivalent simple pendulum

6.3. Damped Oscillatory Motion

In real practice, there would be a number of external resistive forces acting on an oscillating system,

- lowering its energy and
- Decreasing the amplitudes of successive oscillations.

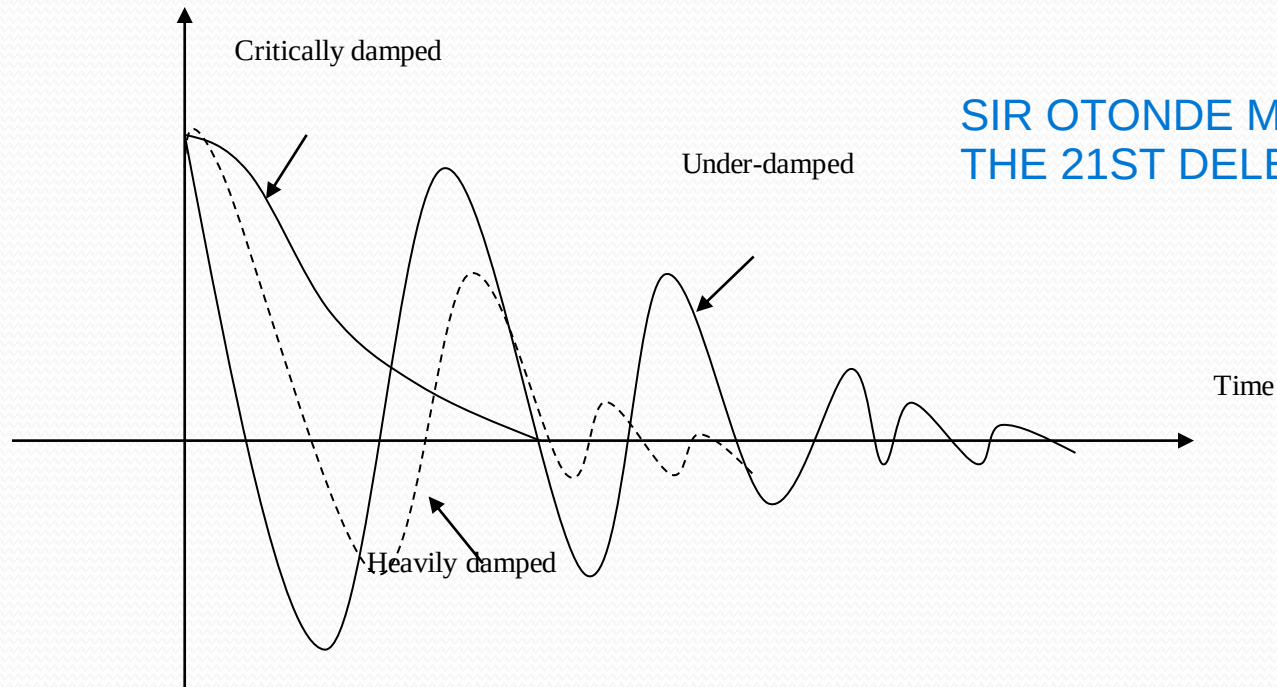
The effects of such forces on the amplitude is known as

damping

the causing forces, like friction/resistance, are called **damping forces**.

The **decaying of amplitude** with time is a characteristic of **damped oscillation**.

- Represented in graphical form as in Fig.



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Damped SHM: the force law

It is important to note that,

- **the damping forces** act against the displacement (like the restoring force)
- Thus, both of them act in the opposite direction and are marked as negative (opposite displacement)

If **in addition to** the restoring force **$F = -kx$** we add a **damping force** ($-F_d$) then we have a general law Newton's 2nd Law,

$$-(kx + F_d) = m \frac{d^2 x}{dt^2}$$

$$-(kx + F_d) = m \frac{d^2 x}{dt^2}$$

- This equation may be written in the form generally known as : **Equation for damped harmonic oscillation.**

$$m \frac{d^2 x}{dt^2} + F_d + kx = 0$$

- The simplest case of a **damping force varies with the velocity, increasing as the velocity increases** but directed in the opposite direction and can be expressed as:

$$F_d \propto v$$

Where $F_d \propto v$

Can be written as $F_d = \beta v = \beta \frac{dx}{dt}$

- where β is a constant called **damping coefficient**.

Hence, the damped SHM Equation is expressed as:

$$m \frac{d^2 x}{dt^2} + \beta \frac{dx}{dt} + kx = 0$$

OR,

$$\frac{d^2 x}{dt^2} + \frac{\beta}{m} \frac{dx}{dt} + \frac{k}{m} x = 0$$

If we recall that $\frac{k}{m} = \omega^2$ and let $\frac{\beta}{m} = 2\gamma$

Then

$$\frac{d^2x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega^2 x = 0$$

- This equation can be solved to determine how the **damping term** γ influences the amplitude of the displacement (x).

$$y'' + (2\gamma)y' + \omega^2 x = 0$$

For small displacements x , we can read as

$$y^2 + (2\gamma)y + \omega^2 = 0$$

$$y^2 + (2\gamma)y + \omega^2 = 0$$

From

$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$y = \frac{-2\gamma \pm \sqrt{(2\gamma)^2 - 4\omega^2}}{2} = -\gamma \pm \frac{\sqrt{(2\gamma)^2 - 4\omega^2}}{2}$$

Solutions depend on the terms in the root

$$\sqrt{4\gamma^2 - 4\omega^2}$$

- Options are
- i) $4\gamma^2 = 4\omega^2$ OR $\gamma = \omega$
 - ii) $4\gamma^2 > 4\omega^2$ OR $\gamma > \omega$
 - iii) $4\gamma^2 = 4\omega^2$ OR $\gamma = \omega$

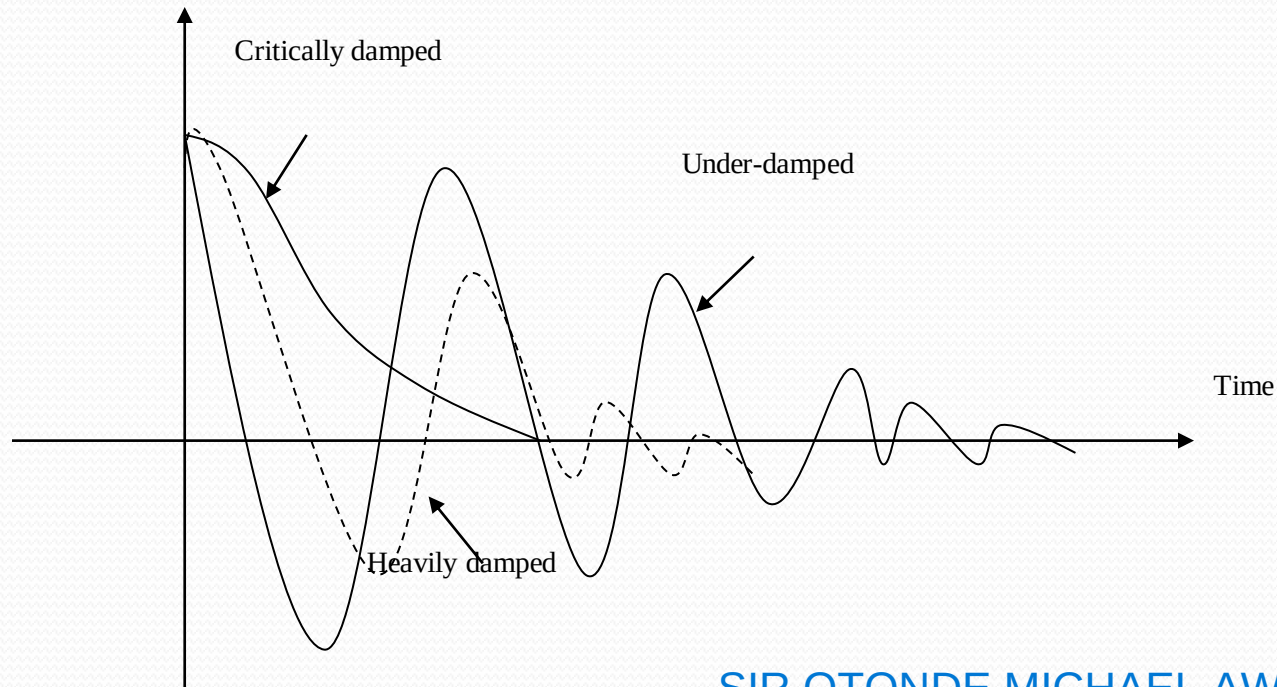
Damping can be categorized, depending on the relation between , ω and γ as:

- Under-damped ($\gamma < \omega$)
- Heavily damped ($\gamma > \omega$)
- Critically damped ($\omega = \gamma$)

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- Represented in graphical form as in Fig.



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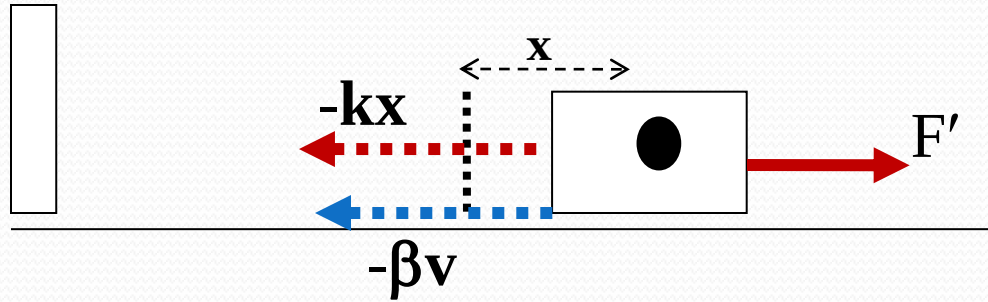
Forced oscillations and Resonance

A number of SHM oscillator are those that **move freely**, sustained by their **natural restoring force** only, like:

- a pendulum and mass on a spring.

However, **natural restoring forces only** never sustain oscillatory motion **due to resistive forces**.

One way to make an oscillator **sustain its amplitude** of oscillation, without decay, is **to provide external force**, in the **direction of displacement**, to support /force motion.



This would require the **supportive force** (F') and the **oscillator** keep pace with each other. Otherwise it would act like another damping force

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i.e, the applied **external force** and the oscillator must oscillate at **the same frequency**, described as **being in resonance**.

Definition:

Resonance is a state of forced oscillation in which the frequency of applied external force and that of the oscillator are equal.

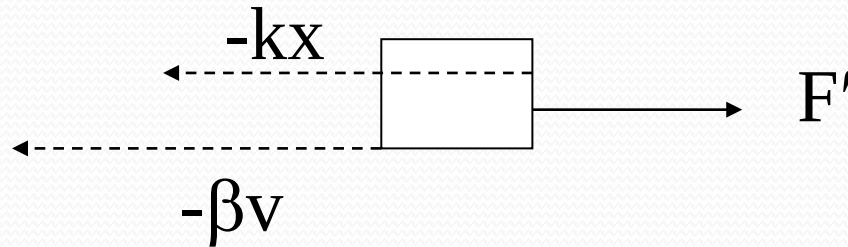
- If the applied external force F' should also be oscillatory then its equation should be that of oscillation,

Say, such as;

$$F' = F_0 \cos \alpha t \text{ or in "Sin".}$$

Where F_0 – is amplitude of the force.
 α – its angular frequency

Then, from



And by Newton's 2nd Law of net force ($F = ma$)

$$F' - kx - \beta v = m \frac{d^2 x}{dt^2}$$

Can be expressed as

$$F' = m \frac{d^2 x}{dt^2} + \beta v + kx$$

Which may be generally written as (Forced oscillation)

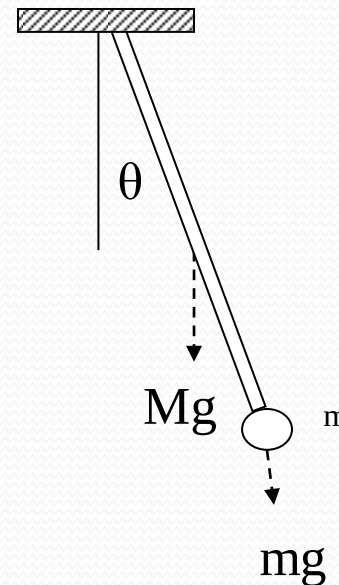
$$m \frac{d^2 x}{dt^2} + \beta \frac{dx}{dt} + kx = F_o \cos at$$

Example

A compound pendulum made of a *thin uniform rod* of mass M and length l is *fixed at one end*, with a *point mass* of mass m at *the other end*. Show that *the expression* for its *angular acceleration* when it is at an angle θ , Fig., is

$$\alpha = \frac{3(M + 2m)g \sin \theta}{2(M + 3m)l}$$

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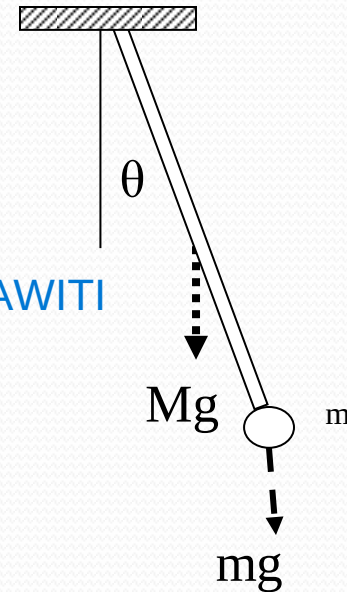
- From Newton's 2nd Law **for rotational motion**

Angular acceleration is given by

Net torque $\tau = I\alpha$

Let $\alpha = \frac{\tau}{I}$

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For a system of connected bodies:

- Net torque is the **sum of the torques** about **an axis**

$$\tau = \tau_r + \tau_m$$

- Moment of inertia is also the **sum of moments** of inertia

$$I = I_r + I_m$$

From Net torque $\tau = \tau_r + \tau_m$

$$\tau_r = r F_{\perp} = \left(\frac{1}{2}l\right) Mg \sin \theta$$

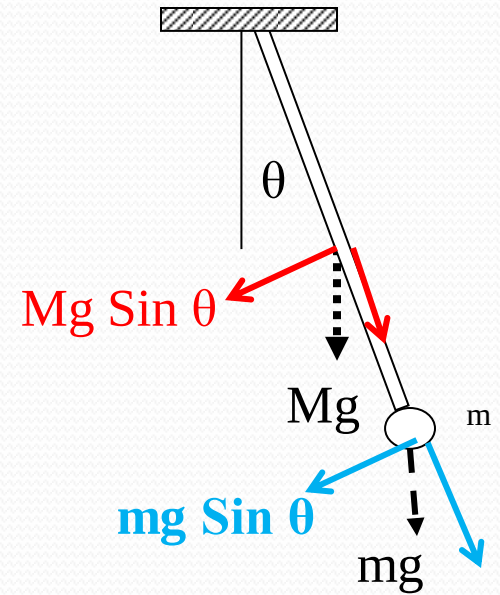
$$\tau_m = r F_{\perp} = lmg \sin \theta$$

Hence, $\tau = \left(\frac{1}{2}l\right) Mg \sin \theta + lmg \sin \theta$

• Also, from moment of inertia $I = I_r + I_m$

$$I_r = \frac{1}{3}Ml^2 \quad \text{and} \quad I_m = ml^2$$

Hence, $I = \frac{1}{3}Ml^2 + ml^2$



$$\tau = \left(\frac{1}{2}l\right) Mg \sin \theta + lmg \sin \theta$$

$$I = \frac{1}{3}Ml^2 + ml^2$$

From $\alpha = \frac{\tau}{I}$

Therefore,
$$\alpha = \frac{\left(\frac{1}{2}l\right) Mg \sin \theta + lmg \sin \theta}{\frac{1}{3}Ml^2 + ml^2}$$

$$= \frac{\left(\frac{1}{2}\right) (M + 2m)lg \sin \theta}{\frac{1}{3} (M + 3m) l^2}$$

$$= \left(\frac{3}{2}\right) \frac{(M + 2m)g \sin \theta}{(M + 3m) l}$$



Exercise

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E-tivity 6.0

Provided in the Module

Ensure :

- you read and understand what to be done.
- You do the stated e-tivities.

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